

RQMC comparison: SmoothPerB4 s = 2 (10<sup>4</sup> samples)

$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

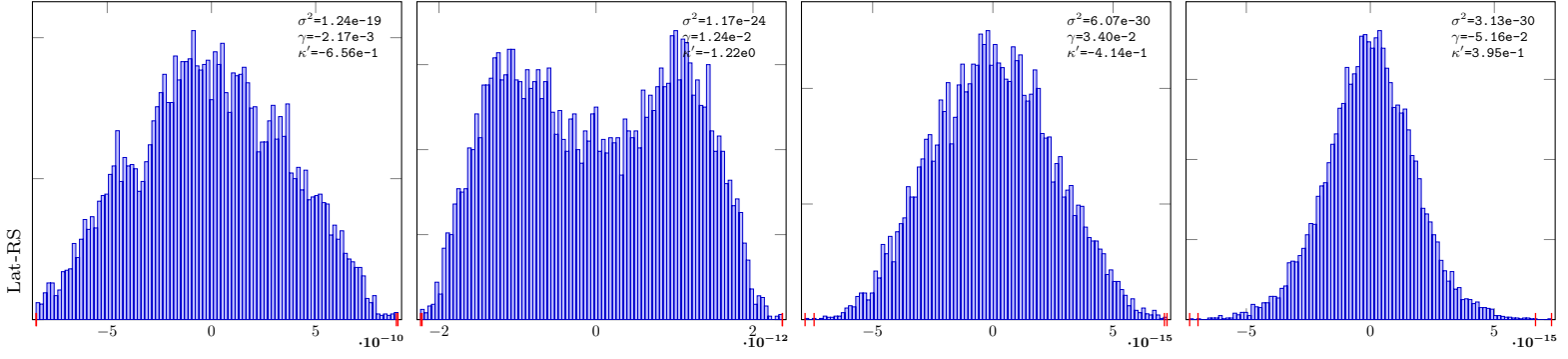
$n = 2^{16}$

SmoothPerB4-2-Lat-RS-10

SmoothPerB4-2-Lat-RS-12

SmoothPerB4-2-Lat-RS-14

SmoothPerB4-2-Lat-RS-16

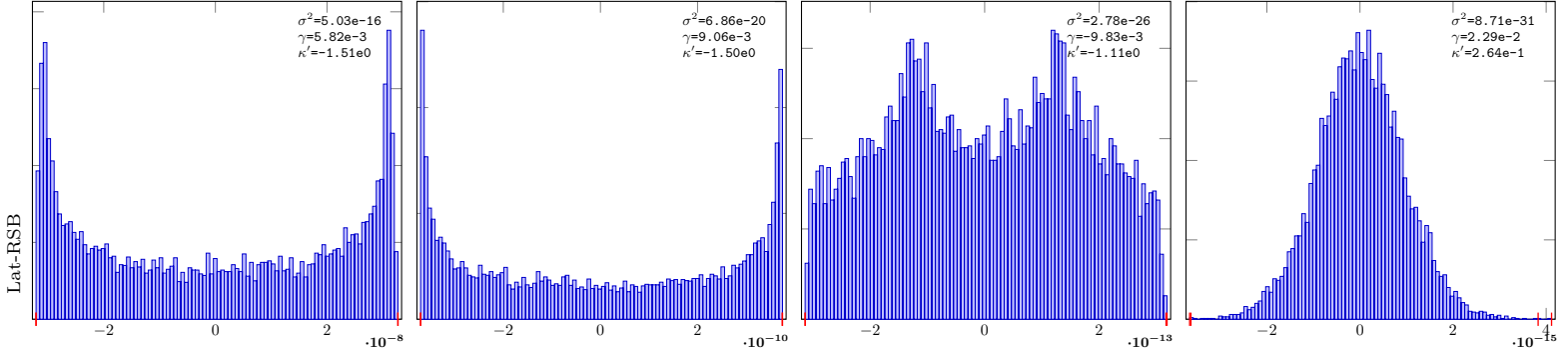


SmoothPerB4-2-Lat-RSB-10

SmoothPerB4-2-Lat-RSB-12

SmoothPerB4-2-Lat-RSB-14

SmoothPerB4-2-Lat-RSB-16

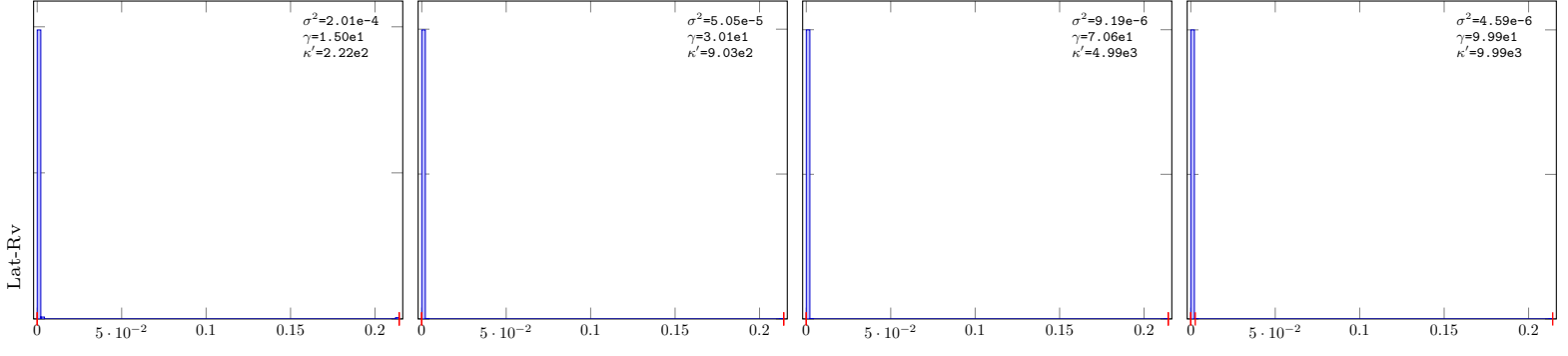


SmoothPerB4-2-Lat-Rv-10

SmoothPerB4-2-Lat-Rv-12

SmoothPerB4-2-Lat-Rv-14

SmoothPerB4-2-Lat-Rv-16

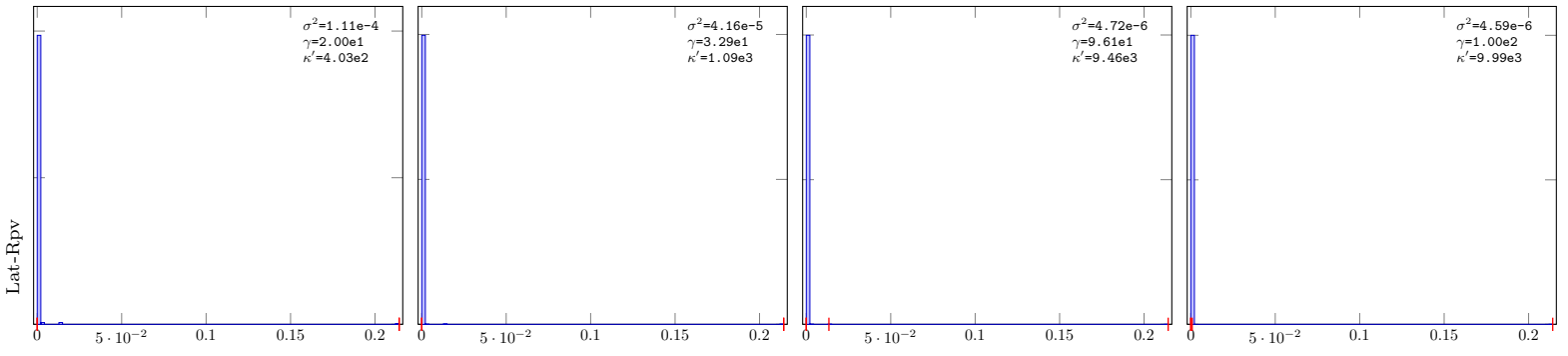


SmoothPerB4-2-Lat-Rpv-10

SmoothPerB4-2-Lat-Rpv-12

SmoothPerB4-2-Lat-Rpv-14

SmoothPerB4-2-Lat-Rpv-16



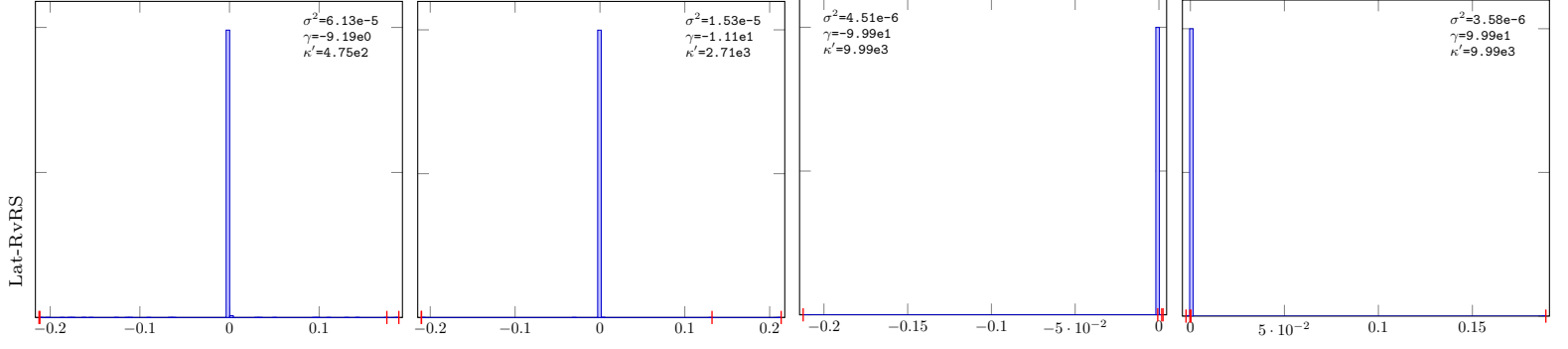
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-2-Lat-RvRS-10

SmoothPerB4-2-Lat-RvRS-12

SmoothPerB4-2-Lat-RvRS-14

SmoothPerB4-2-Lat-RvRS-16

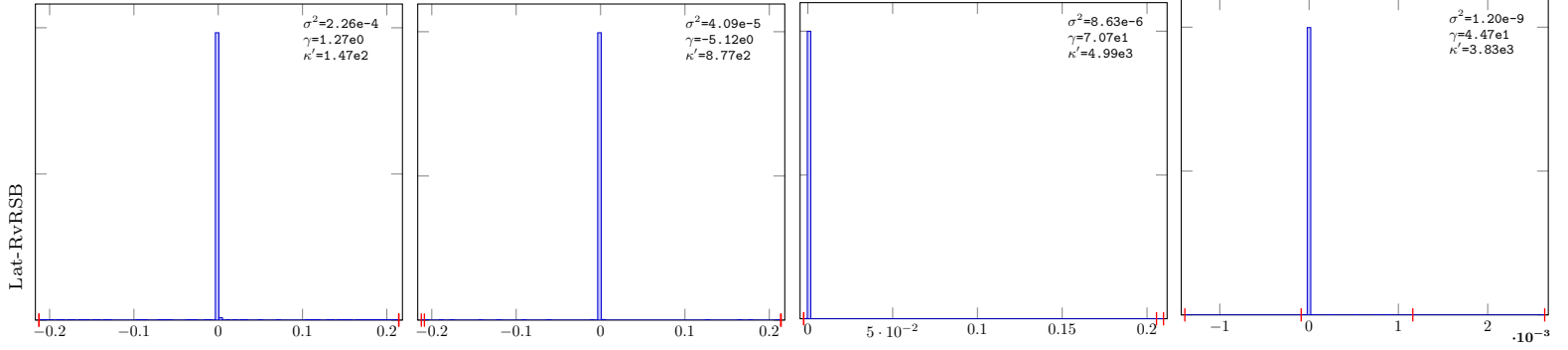


SmoothPerB4-2-Lat-RvRSB-10

SmoothPerB4-2-Lat-RvRSB-12

SmoothPerB4-2-Lat-RvRSB-14

SmoothPerB4-2-Lat-RvRSB-16

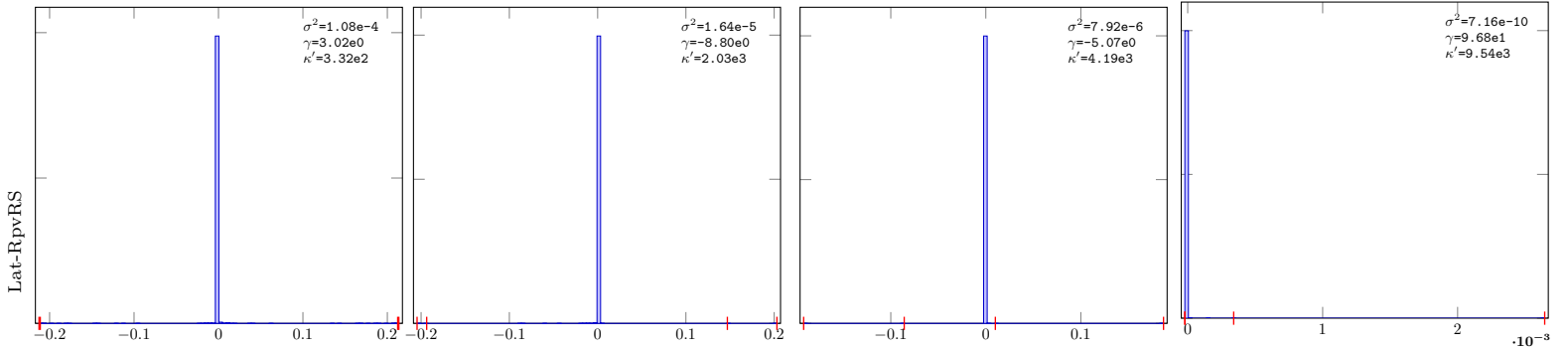


SmoothPerB4-2-Lat-RpvRS-10

SmoothPerB4-2-Lat-RpvRS-12

SmoothPerB4-2-Lat-RpvRS-14

SmoothPerB4-2-Lat-RpvRS-16

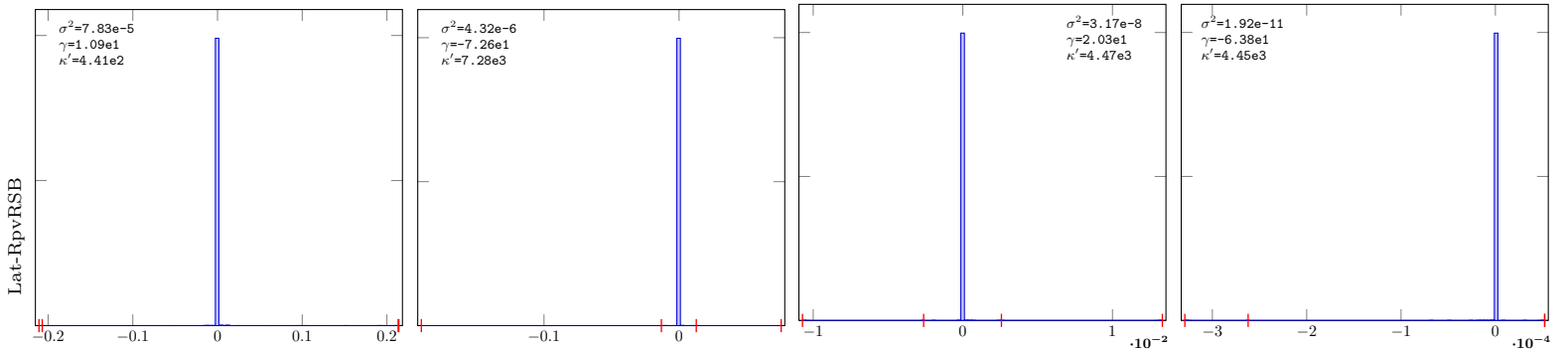


SmoothPerB4-2-Lat-RpvRSB-10

SmoothPerB4-2-Lat-RpvRSB-12

SmoothPerB4-2-Lat-RpvRSB-14

SmoothPerB4-2-Lat-RpvRSB-16



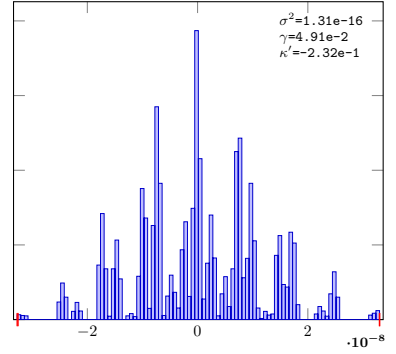
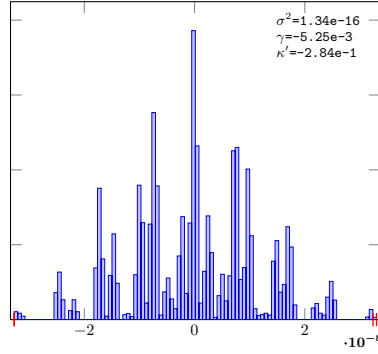
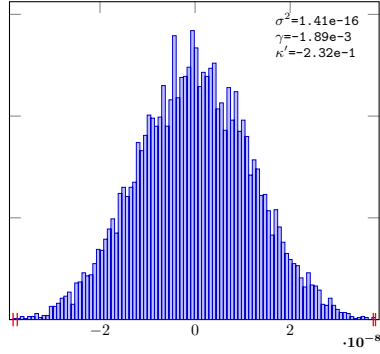
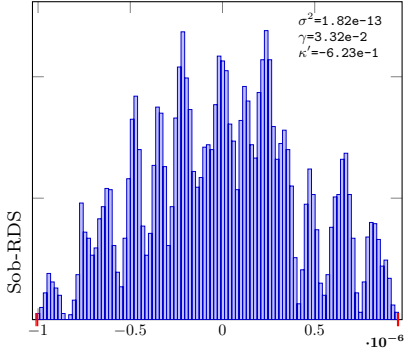
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-2-Sob-RDS-10

SmoothPerB4-2-Sob-RDS-12

SmoothPerB4-2-Sob-RDS-14

SmoothPerB4-2-Sob-RDS-16

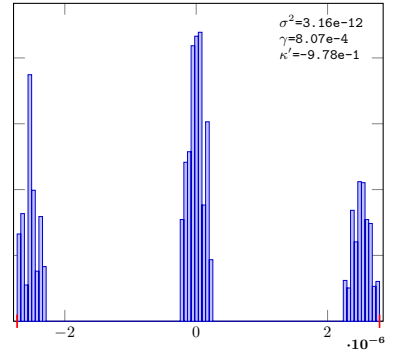
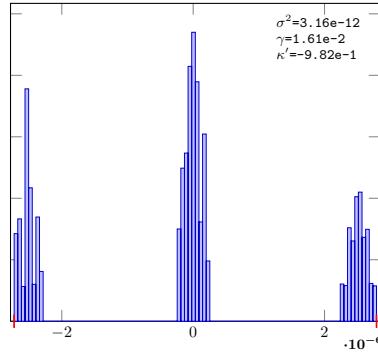
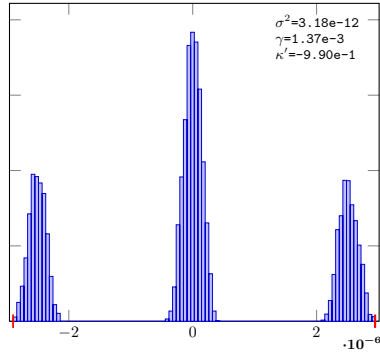
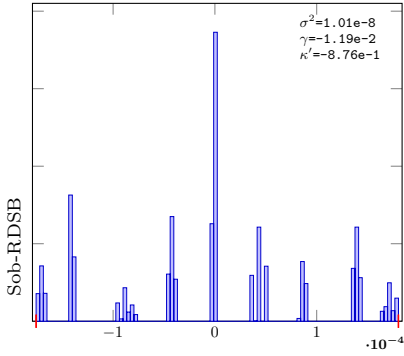


SmoothPerB4-2-Sob-RDSB-10

SmoothPerB4-2-Sob-RDSB-12

SmoothPerB4-2-Sob-RDSB-14

SmoothPerB4-2-Sob-RDSB-16

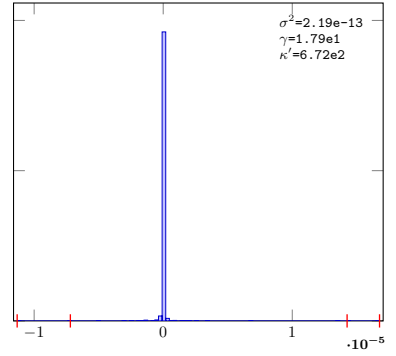
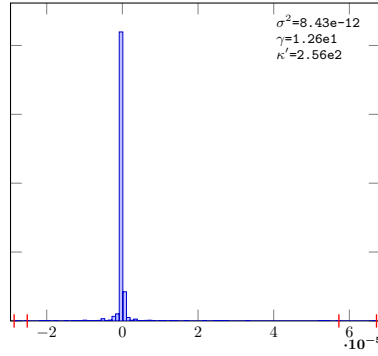
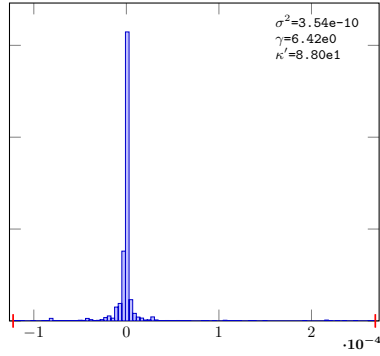
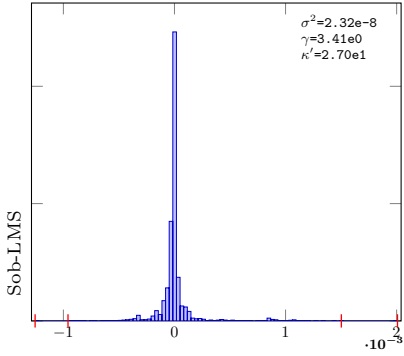


SmoothPerB4-2-Sob-LMS-10

SmoothPerB4-2-Sob-LMS-12

SmoothPerB4-2-Sob-LMS-14

SmoothPerB4-2-Sob-LMS-16

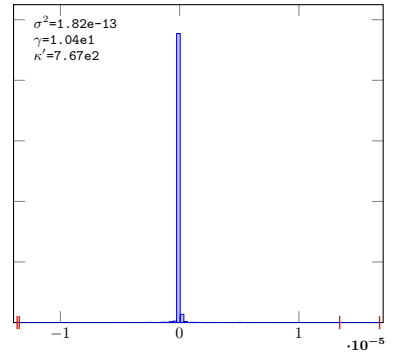
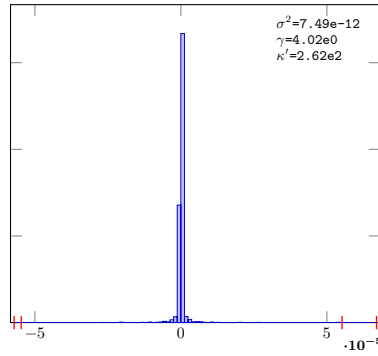
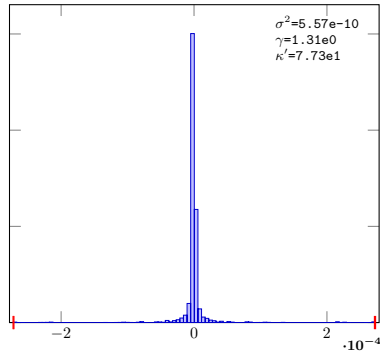
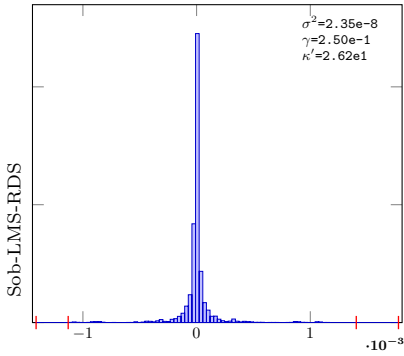


SmoothPerB4-2-Sob-LMS-RDS-10

SmoothPerB4-2-Sob-LMS-RDS-12

SmoothPerB4-2-Sob-LMS-RDS-14

SmoothPerB4-2-Sob-LMS-RDS-16



$$n = 2^{10}$$

$$n = 2^{12}$$

$$n = 2^{14}$$

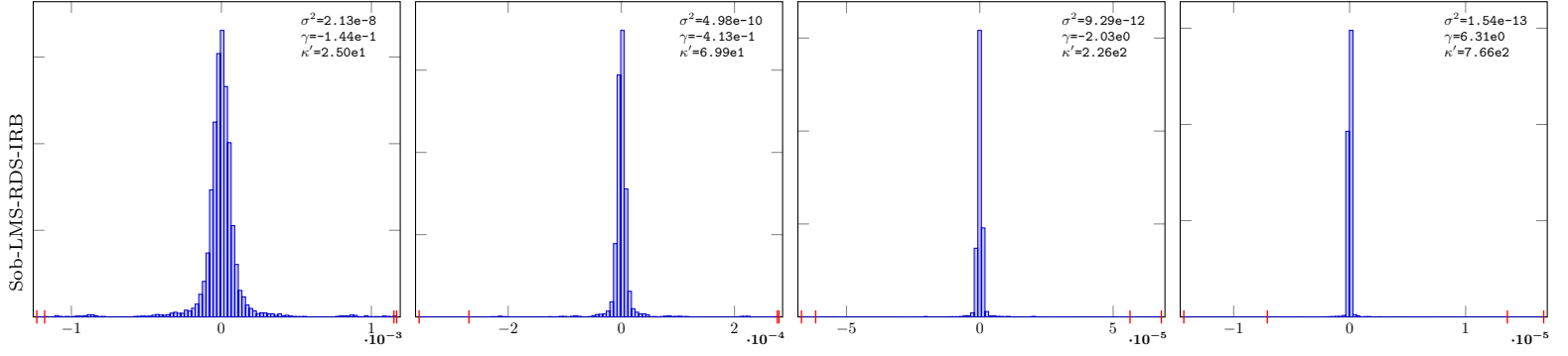
$$n = 2^{16}$$

SmoothPerB4-2-Sob-LMS-RDS-IRB-10

SmoothPerB4-2-Sob-LMS-RDS-IRB-12

SmoothPerB4-2-Sob-LMS-RDS-IRB-14

SmoothPerB4-2-Sob-LMS-RDS-IRB-16

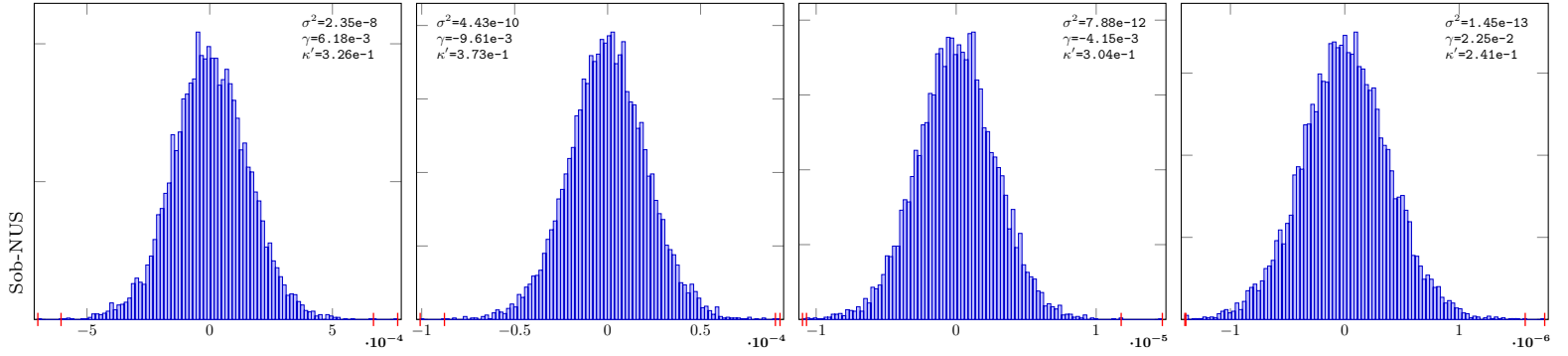


SmoothPerB4-2-Sob-NUS-10

SmoothPerB4-2-Sob-NUS-12

SmoothPerB4-2-Sob-NUS-14

SmoothPerB4-2-Sob-NUS-16



# RQMC comparison: SmoothPerB4 s = 4 (10<sup>4</sup> samples)

$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

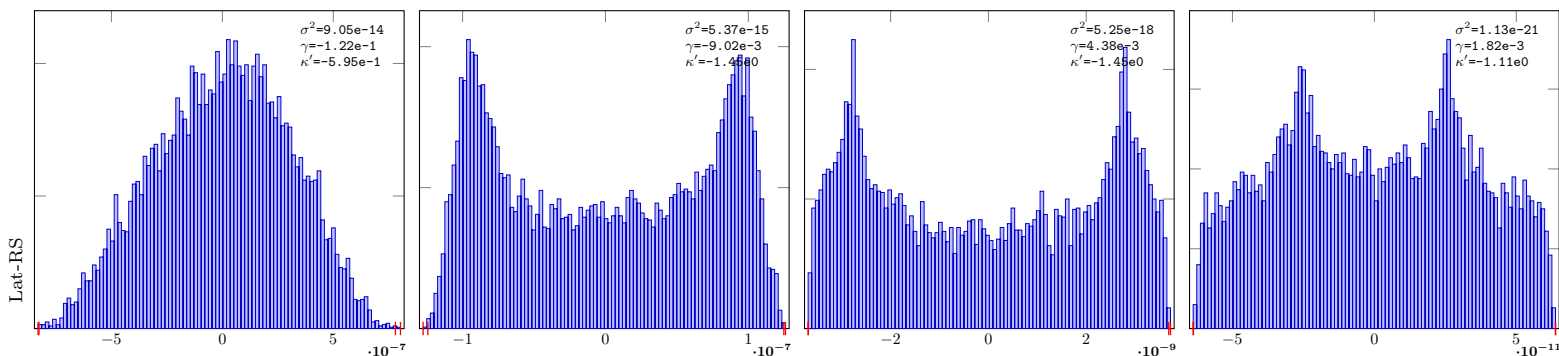
$n = 2^{16}$

SmoothPerB4-4-Lat-RS-10

SmoothPerB4-4-Lat-RS-12

SmoothPerB4-4-Lat-RS-14

SmoothPerB4-4-Lat-RS-16

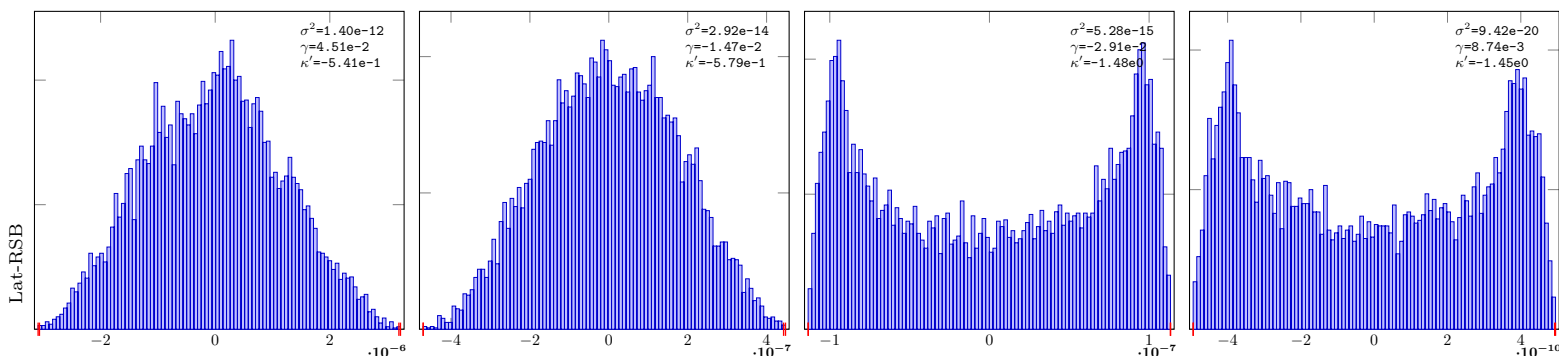


SmoothPerB4-4-Lat-RSB-10

SmoothPerB4-4-Lat-RSB-12

SmoothPerB4-4-Lat-RSB-14

SmoothPerB4-4-Lat-RSB-16

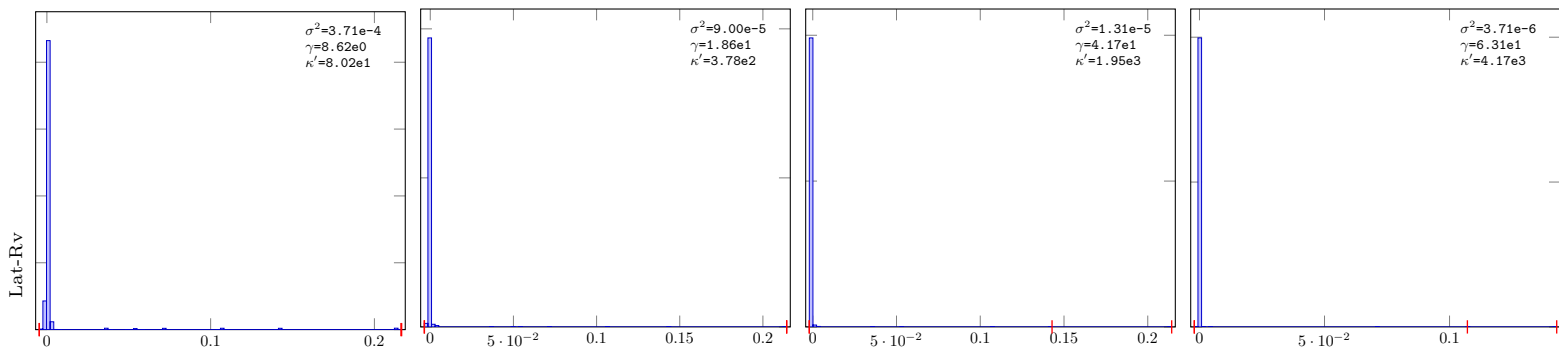


SmoothPerB4-4-Lat-Rv-10

SmoothPerB4-4-Lat-Rv-12

SmoothPerB4-4-Lat-Rv-14

SmoothPerB4-4-Lat-Rv-16

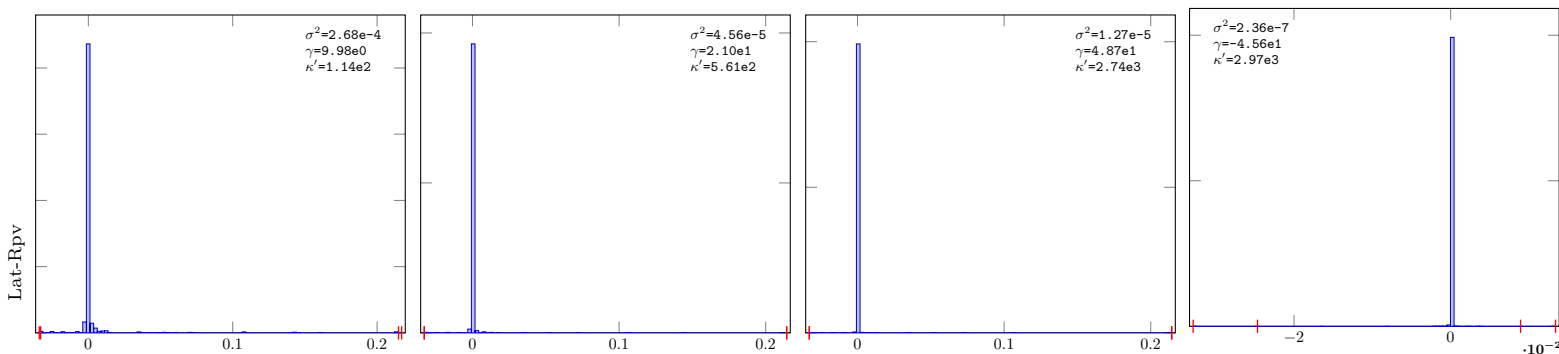


SmoothPerB4-4-Lat-Rpv-10

SmoothPerB4-4-Lat-Rpv-12

SmoothPerB4-4-Lat-Rpv-14

SmoothPerB4-4-Lat-Rpv-16



$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

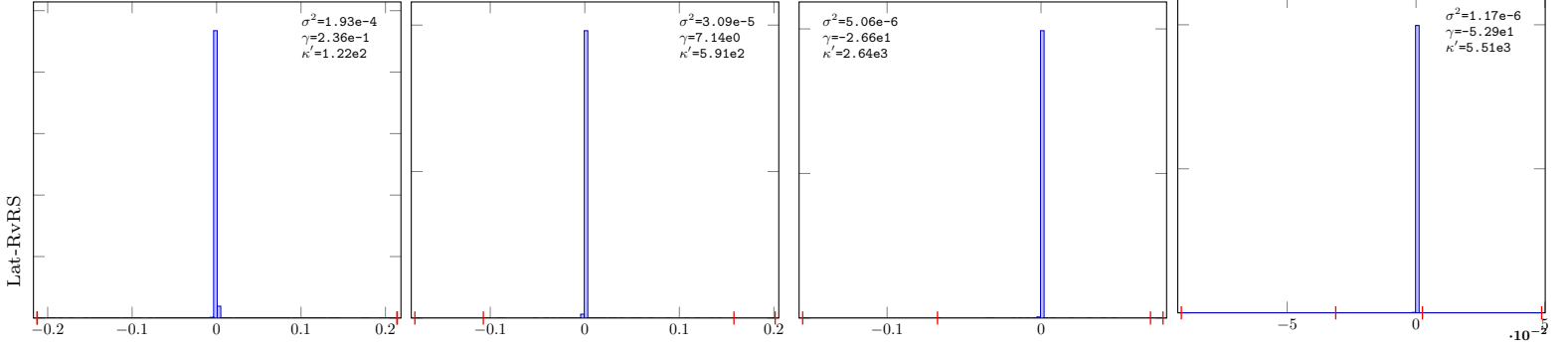
$n = 2^{16}$

SmoothPerB4-4-Lat-RvRS-10

SmoothPerB4-4-Lat-RvRS-12

SmoothPerB4-4-Lat-RvRS-14

SmoothPerB4-4-Lat-RvRS-16

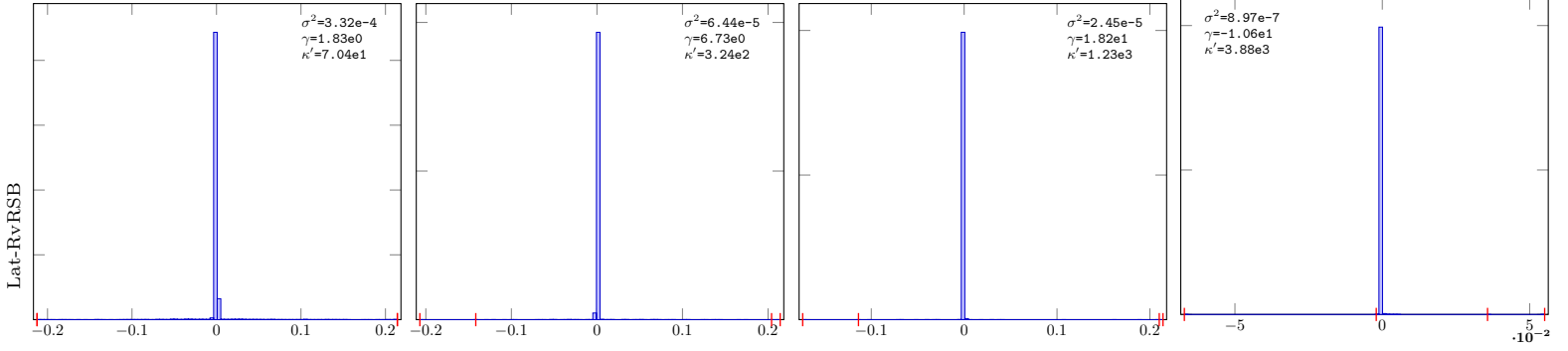


SmoothPerB4-4-Lat-RvRSB-10

SmoothPerB4-4-Lat-RvRSB-12

SmoothPerB4-4-Lat-RvRSB-14

SmoothPerB4-4-Lat-RvRSB-16

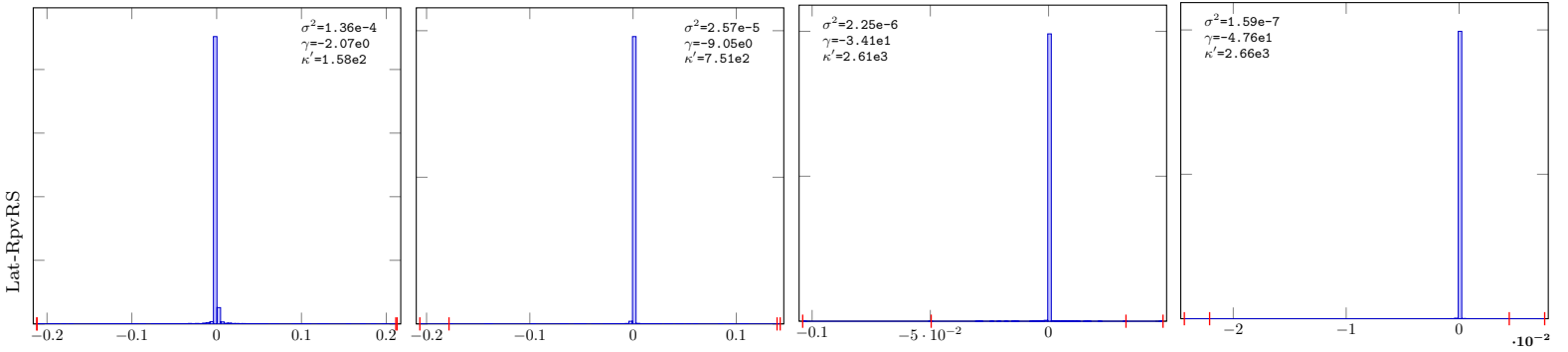


SmoothPerB4-4-Lat-RpvRS-10

SmoothPerB4-4-Lat-RpvRS-12

SmoothPerB4-4-Lat-RpvRS-14

SmoothPerB4-4-Lat-RpvRS-16

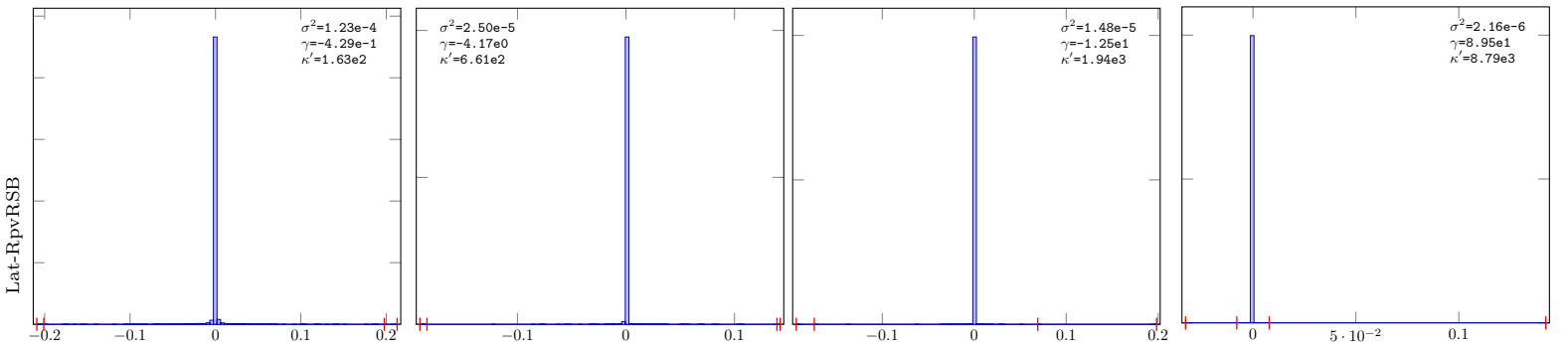


SmoothPerB4-4-Lat-RpvRSB-10

SmoothPerB4-4-Lat-RpvRSB-12

SmoothPerB4-4-Lat-RpvRSB-14

SmoothPerB4-4-Lat-RpvRSB-16



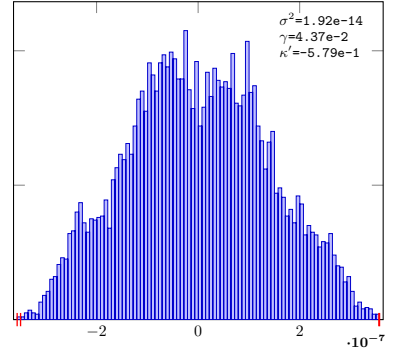
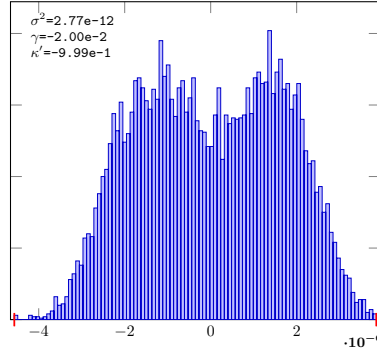
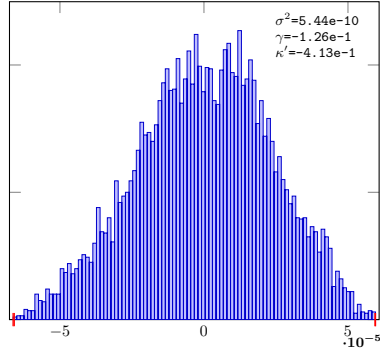
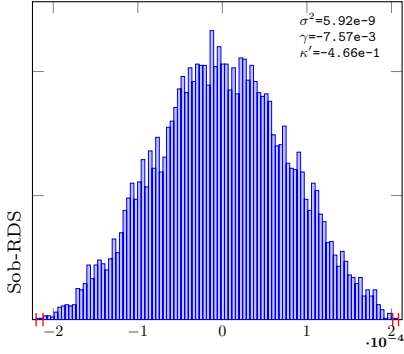
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-4-Sob-RDS-10

SmoothPerB4-4-Sob-RDS-12

SmoothPerB4-4-Sob-RDS-14

SmoothPerB4-4-Sob-RDS-16

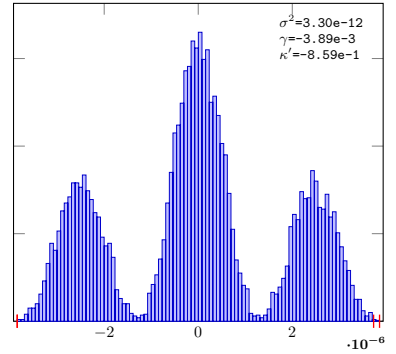
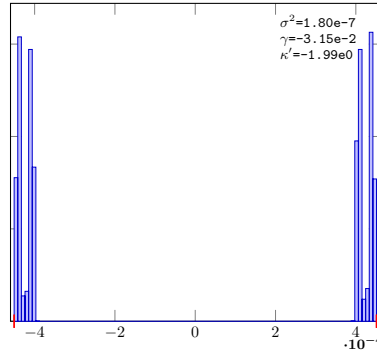
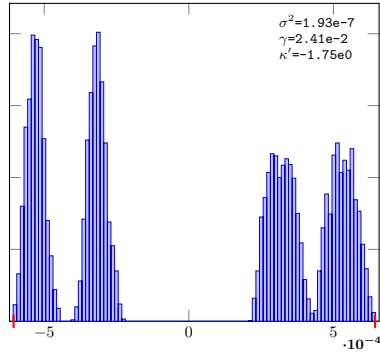
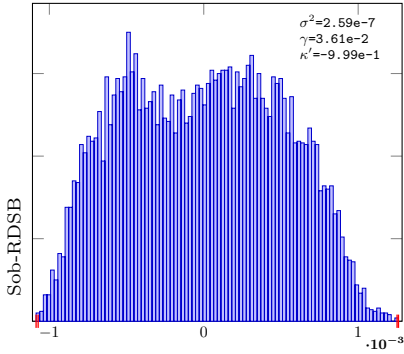


SmoothPerB4-4-Sob-RDSB-10

SmoothPerB4-4-Sob-RDSB-12

SmoothPerB4-4-Sob-RDSB-14

SmoothPerB4-4-Sob-RDSB-16

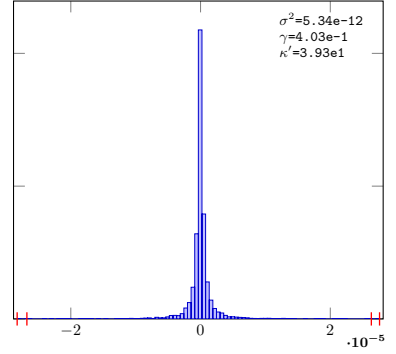
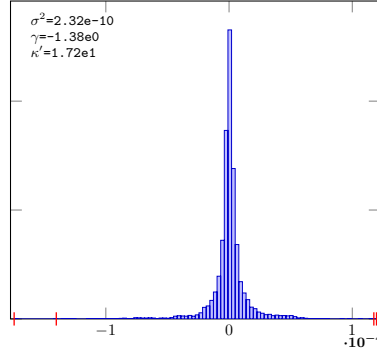
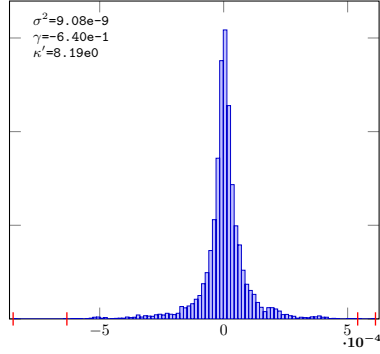
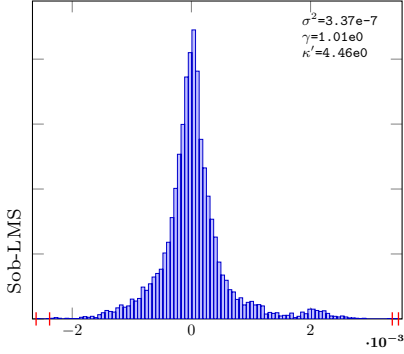


SmoothPerB4-4-Sob-LMS-10

SmoothPerB4-4-Sob-LMS-12

SmoothPerB4-4-Sob-LMS-14

SmoothPerB4-4-Sob-LMS-16

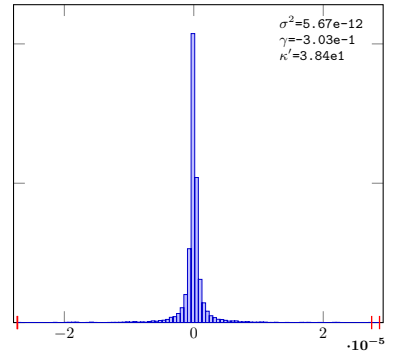
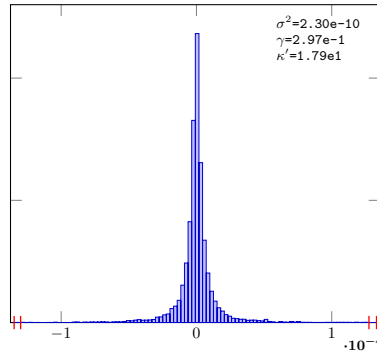
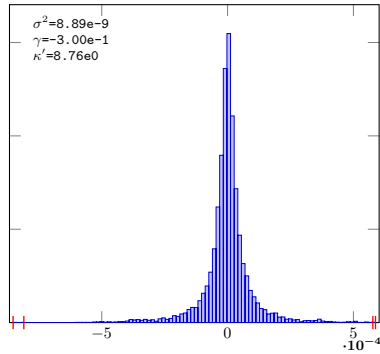
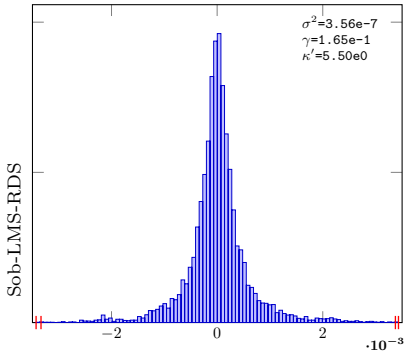


SmoothPerB4-4-Sob-LMS-RDS-10

SmoothPerB4-4-Sob-LMS-RDS-12

SmoothPerB4-4-Sob-LMS-RDS-14

SmoothPerB4-4-Sob-LMS-RDS-16



$$n = 2^{10}$$

$$n = 2^{12}$$

$$n = 2^{14}$$

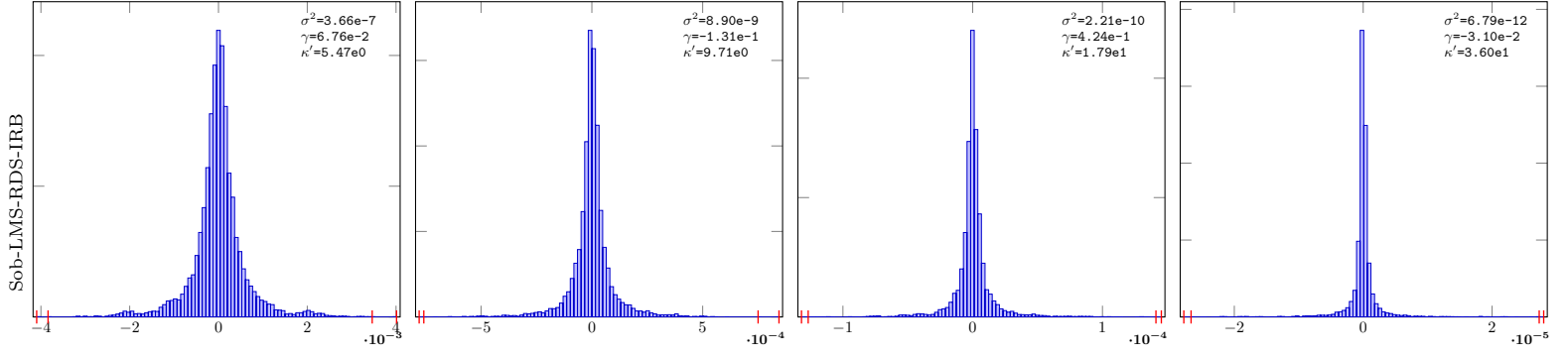
$$n = 2^{16}$$

SmoothPerB4-4-Sob-LMS-RDS-IRB-10

SmoothPerB4-4-Sob-LMS-RDS-IRB-12

SmoothPerB4-4-Sob-LMS-RDS-IRB-14

SmoothPerB4-4-Sob-LMS-RDS-IRB-16

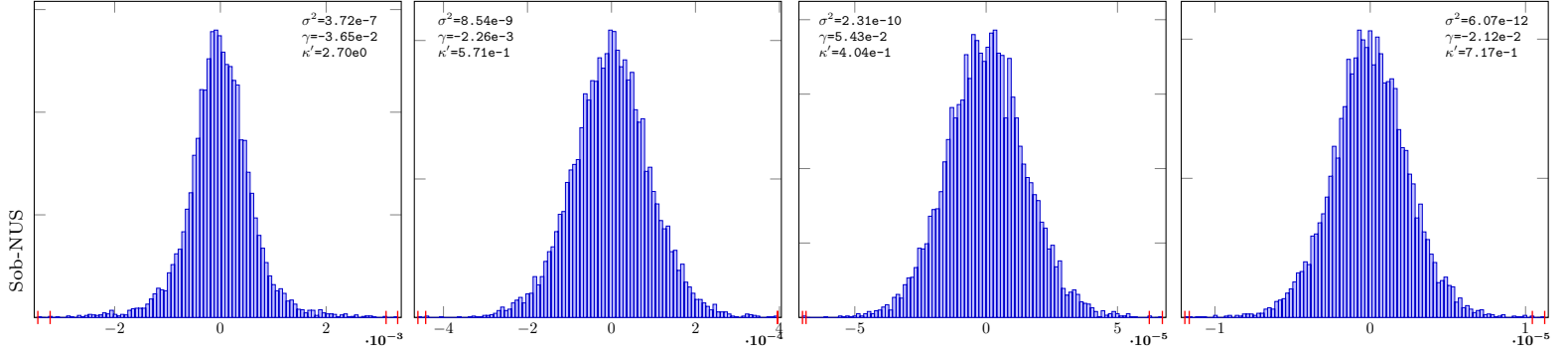


SmoothPerB4-4-Sob-NUS-10

SmoothPerB4-4-Sob-NUS-12

SmoothPerB4-4-Sob-NUS-14

SmoothPerB4-4-Sob-NUS-16



# RQMC comparison: SmoothPerB4 s = 8 (10<sup>4</sup> samples)

$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

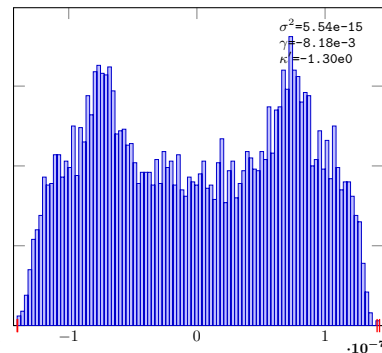
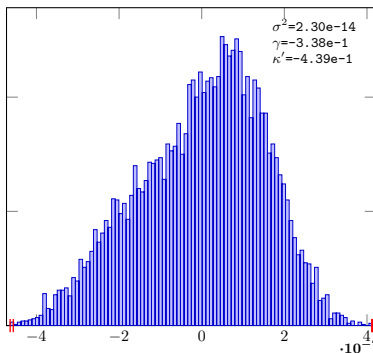
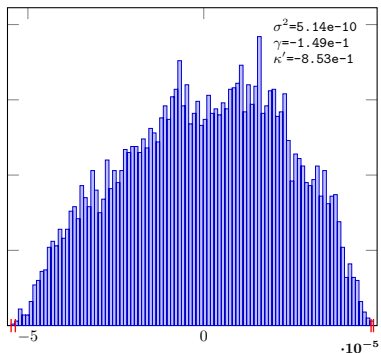
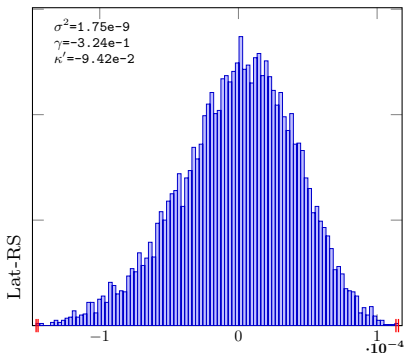
$n = 2^{16}$

SmoothPerB4-8-Lat-RS-10

SmoothPerB4-8-Lat-RS-12

SmoothPerB4-8-Lat-RS-14

SmoothPerB4-8-Lat-RS-16

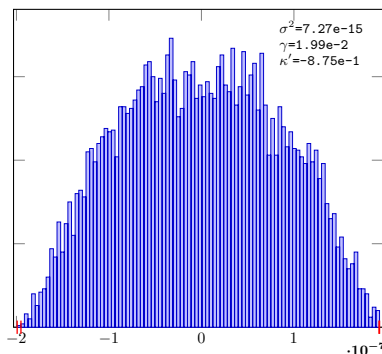
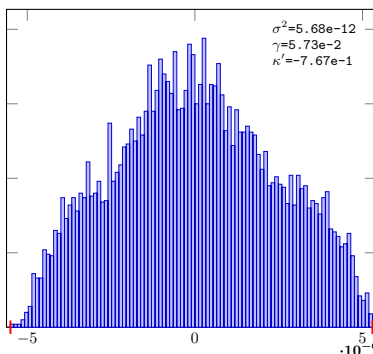
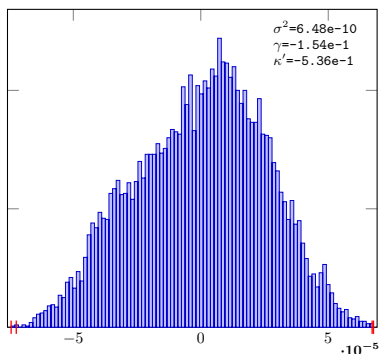
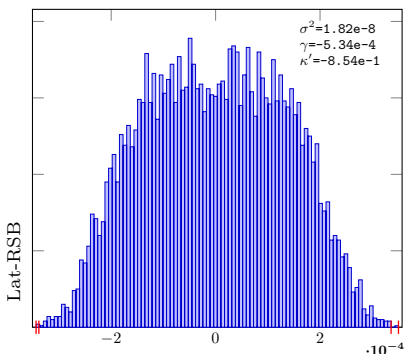


SmoothPerB4-8-Lat-RSB-10

SmoothPerB4-8-Lat-RSB-12

SmoothPerB4-8-Lat-RSB-14

SmoothPerB4-8-Lat-RSB-16

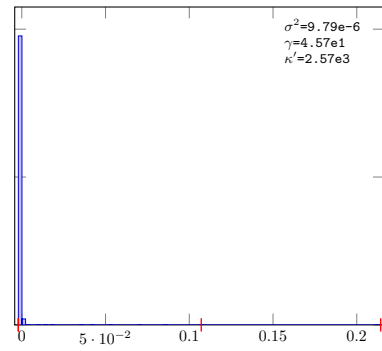
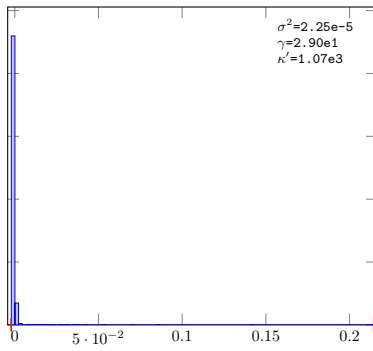
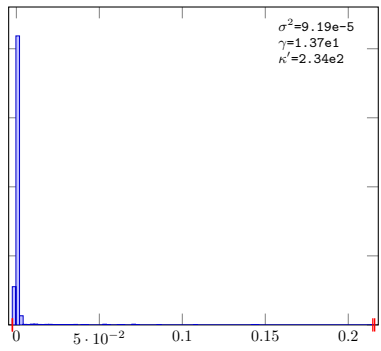
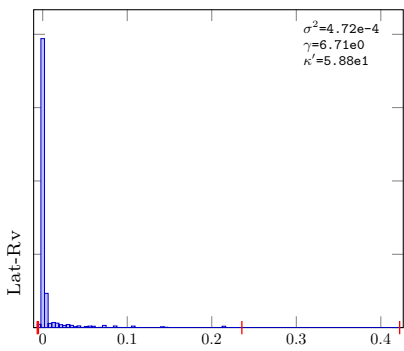


SmoothPerB4-8-Lat-Rv-10

SmoothPerB4-8-Lat-Rv-12

SmoothPerB4-8-Lat-Rv-14

SmoothPerB4-8-Lat-Rv-16

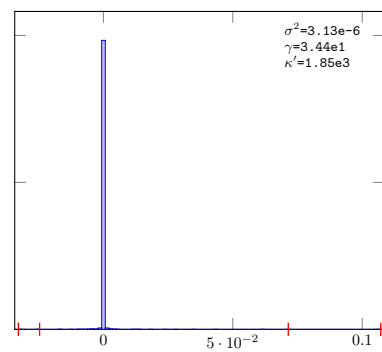
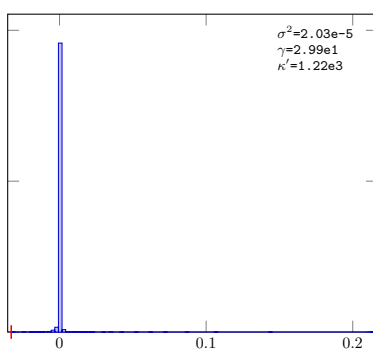
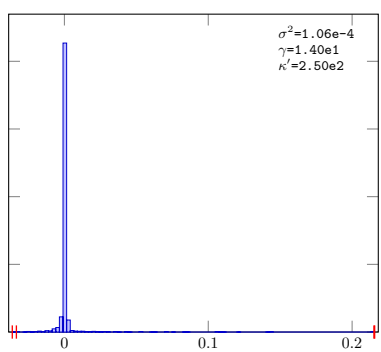
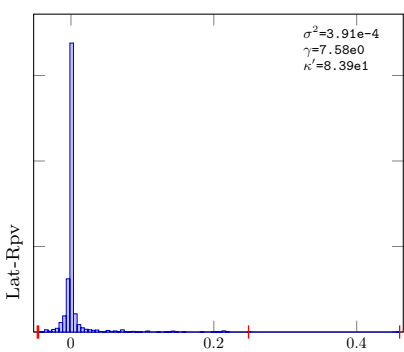


SmoothPerB4-8-Lat-Rpv-10

SmoothPerB4-8-Lat-Rpv-12

SmoothPerB4-8-Lat-Rpv-14

SmoothPerB4-8-Lat-Rpv-16



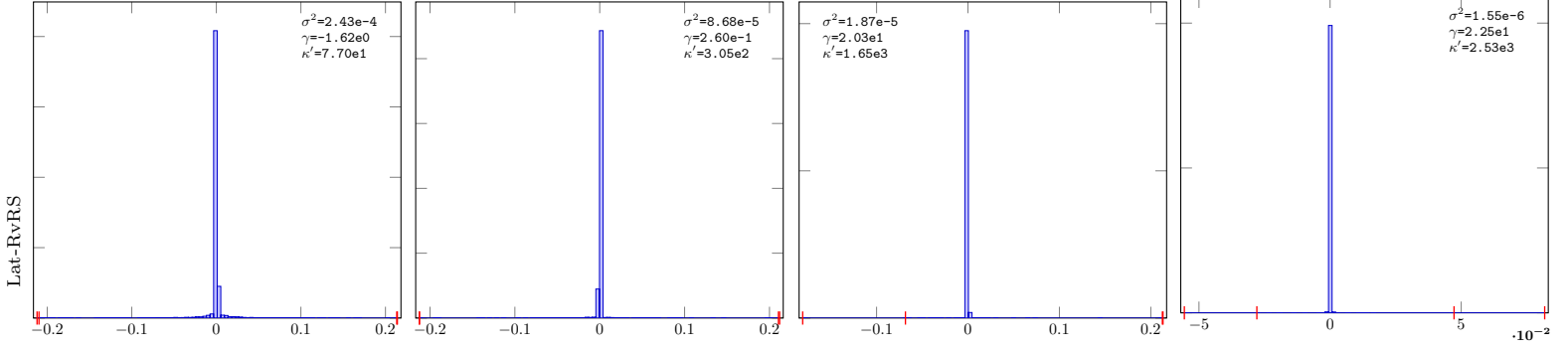
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-8-Lat-RvRS-10

SmoothPerB4-8-Lat-RvRS-12

SmoothPerB4-8-Lat-RvRS-14

SmoothPerB4-8-Lat-RvRS-16

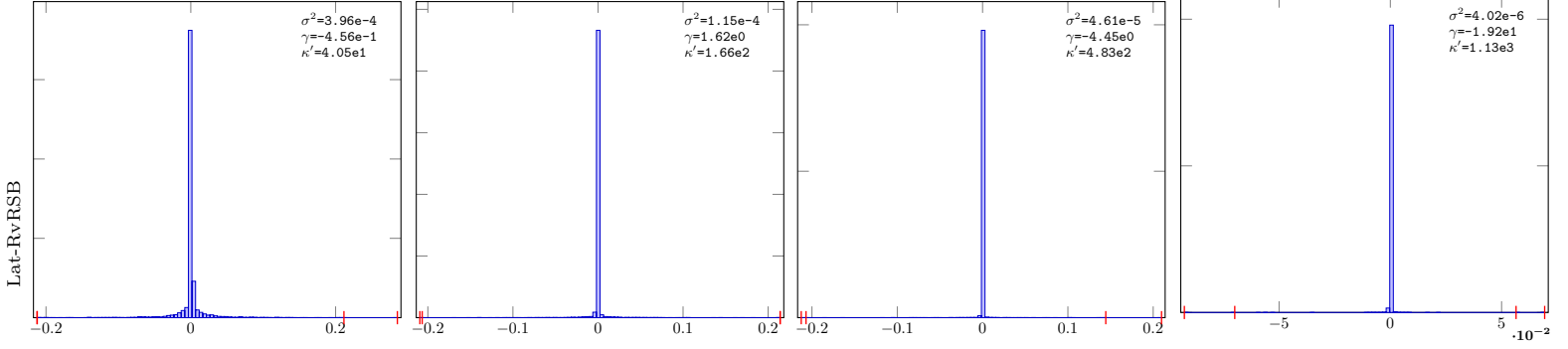


SmoothPerB4-8-Lat-RvRSB-10

SmoothPerB4-8-Lat-RvRSB-12

SmoothPerB4-8-Lat-RvRSB-14

SmoothPerB4-8-Lat-RvRSB-16

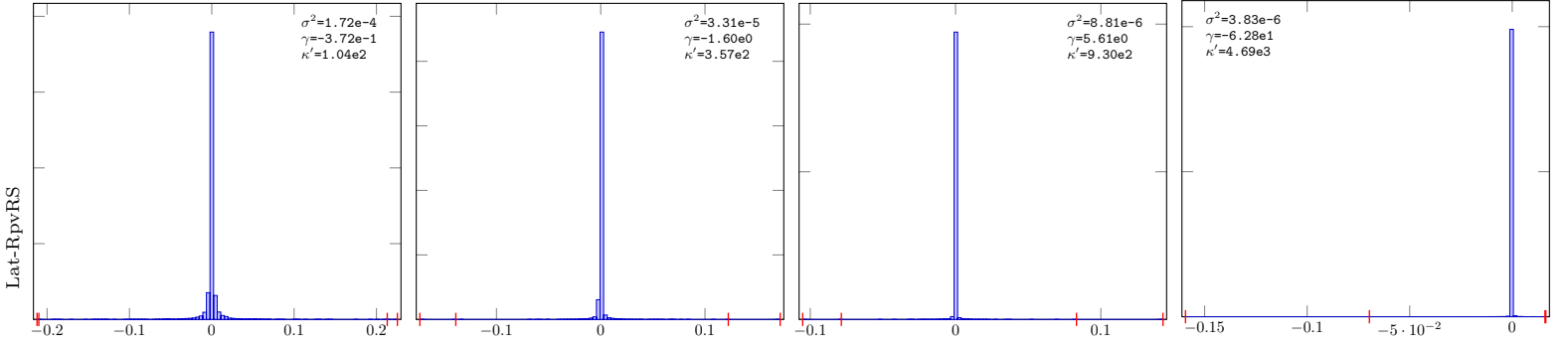


SmoothPerB4-8-Lat-RpvRS-10

SmoothPerB4-8-Lat-RpvRS-12

SmoothPerB4-8-Lat-RpvRS-14

SmoothPerB4-8-Lat-RpvRS-16

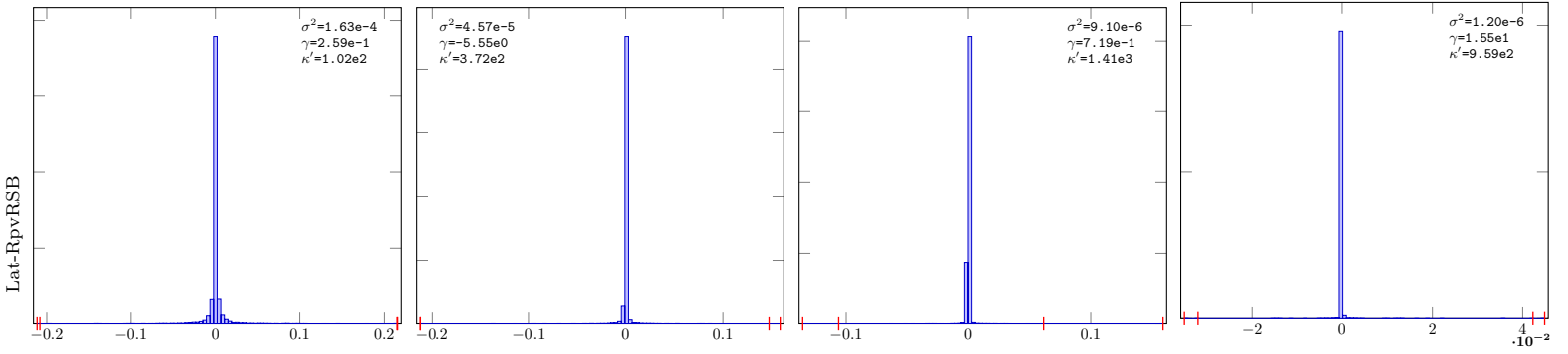


SmoothPerB4-8-Lat-RpvRSB-10

SmoothPerB4-8-Lat-RpvRSB-12

SmoothPerB4-8-Lat-RpvRSB-14

SmoothPerB4-8-Lat-RpvRSB-16



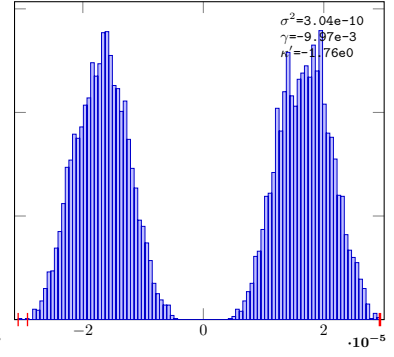
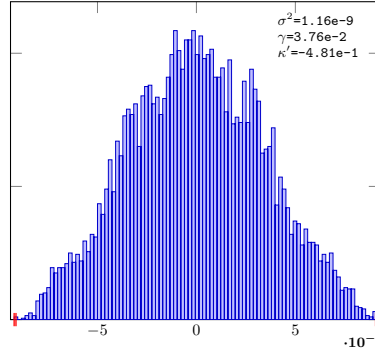
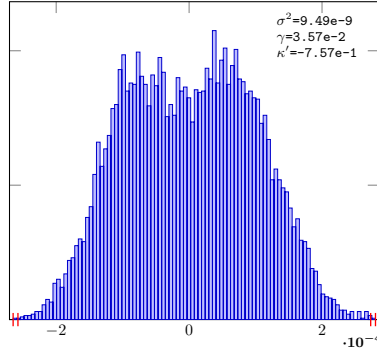
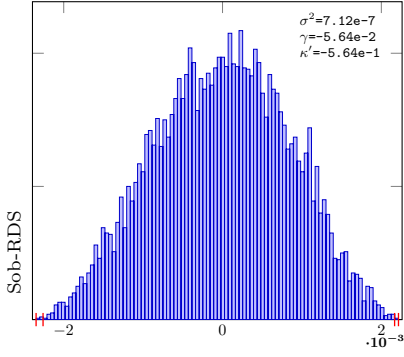
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-8-Sob-RDS-10

SmoothPerB4-8-Sob-RDS-12

SmoothPerB4-8-Sob-RDS-14

SmoothPerB4-8-Sob-RDS-16

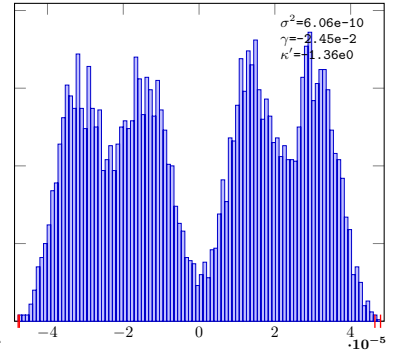
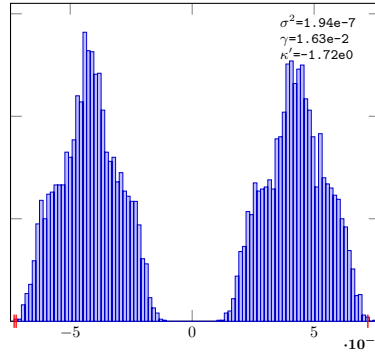
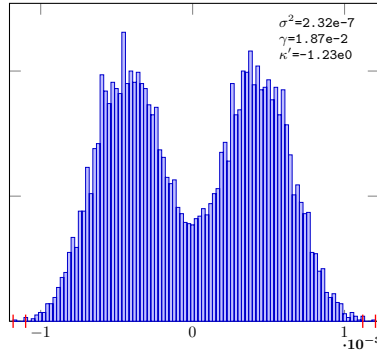
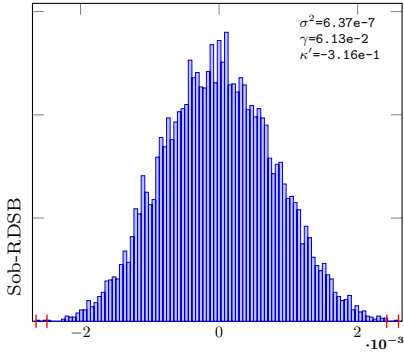


SmoothPerB4-8-Sob-RDSB-10

SmoothPerB4-8-Sob-RDSB-12

SmoothPerB4-8-Sob-RDSB-14

SmoothPerB4-8-Sob-RDSB-16

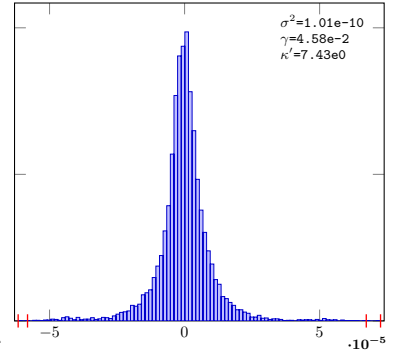
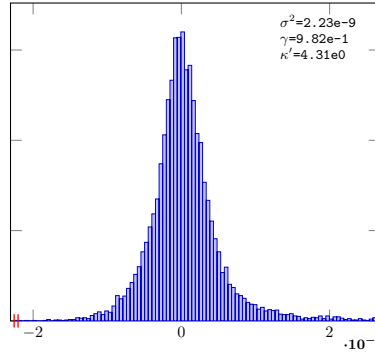
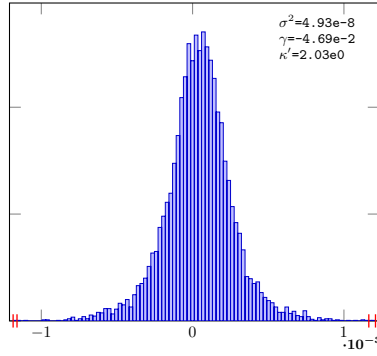
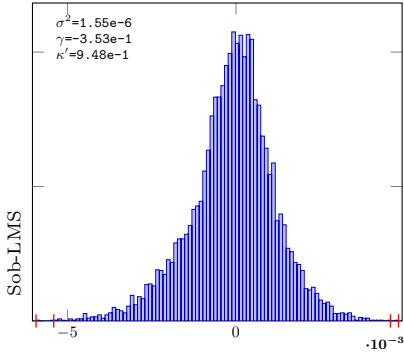


SmoothPerB4-8-Sob-LMS-10

SmoothPerB4-8-Sob-LMS-12

SmoothPerB4-8-Sob-LMS-14

SmoothPerB4-8-Sob-LMS-16

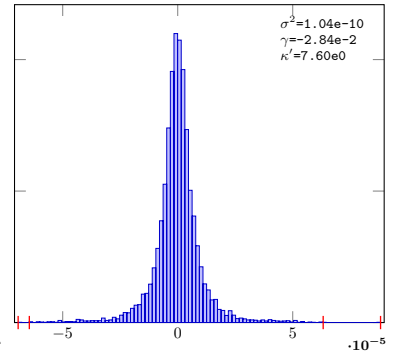
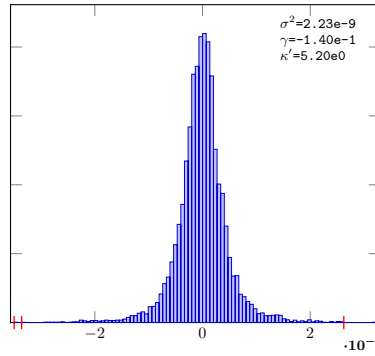
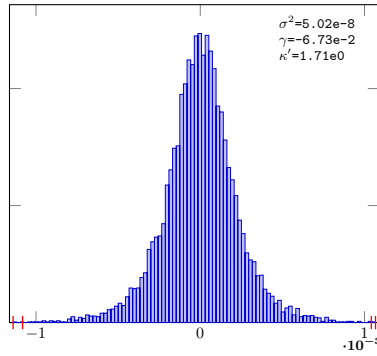
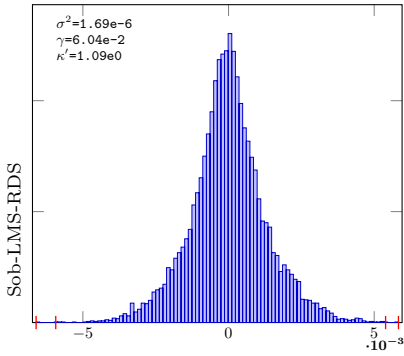


SmoothPerB4-8-Sob-LMS-RDS-10

SmoothPerB4-8-Sob-LMS-RDS-12

SmoothPerB4-8-Sob-LMS-RDS-14

SmoothPerB4-8-Sob-LMS-RDS-16



$$n = 2^{10}$$

$$n = 2^{12}$$

$$n = 2^{14}$$

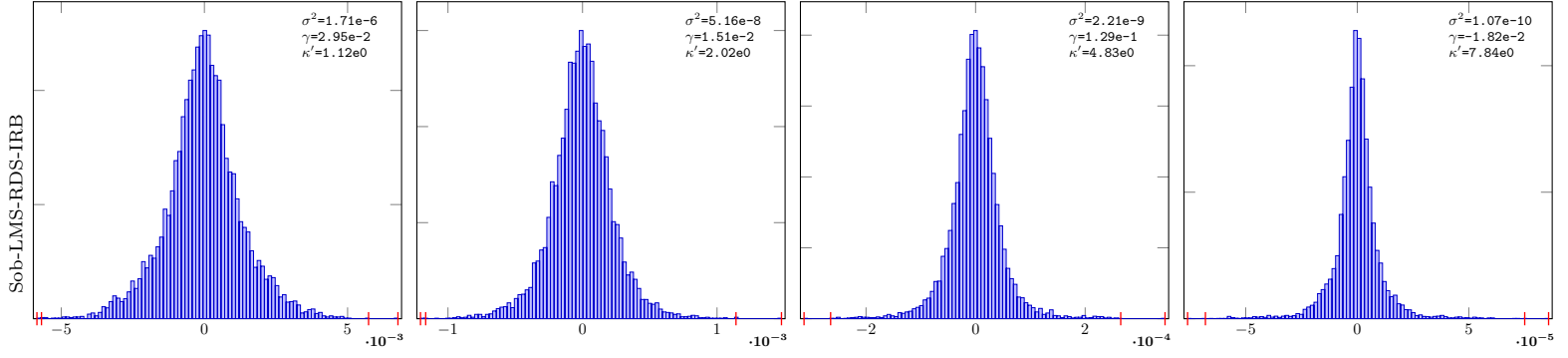
$$n = 2^{16}$$

SmoothPerB4-8-Sob-LMS-RDS-IRB-10

SmoothPerB4-8-Sob-LMS-RDS-IRB-12

SmoothPerB4-8-Sob-LMS-RDS-IRB-14

SmoothPerB4-8-Sob-LMS-RDS-IRB-16

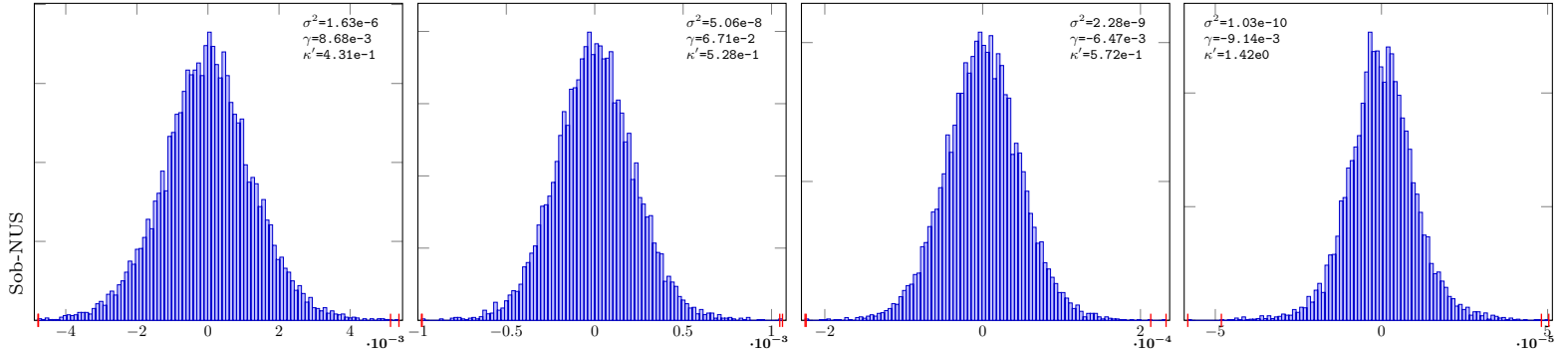


SmoothPerB4-8-Sob-NUS-10

SmoothPerB4-8-Sob-NUS-12

SmoothPerB4-8-Sob-NUS-14

SmoothPerB4-8-Sob-NUS-16



# RQMC comparison: SmoothPerB4 s = 16 ( $10^4$ samples)

$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

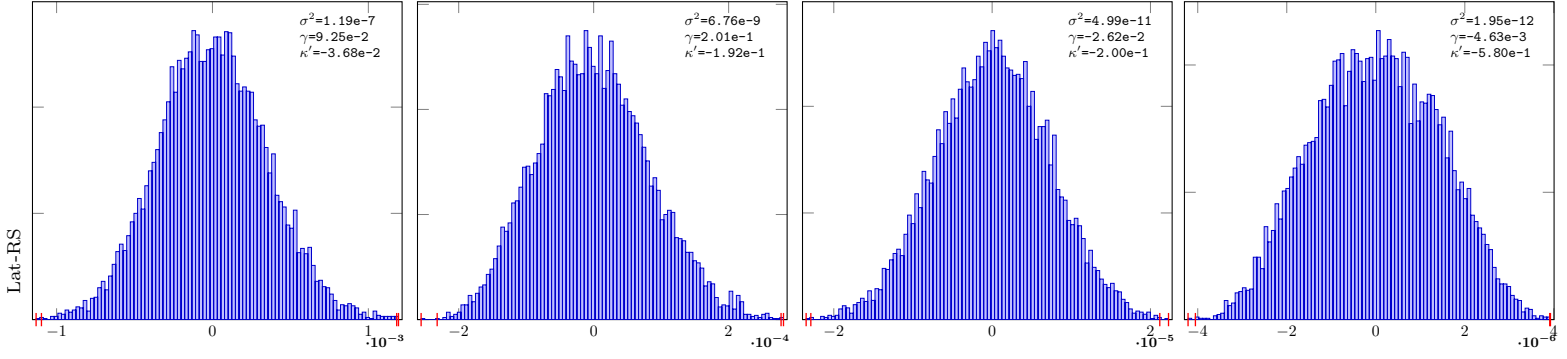
$n = 2^{16}$

SmoothPerB4-16-Lat-RS-10

SmoothPerB4-16-Lat-RS-12

SmoothPerB4-16-Lat-RS-14

SmoothPerB4-16-Lat-RS-16

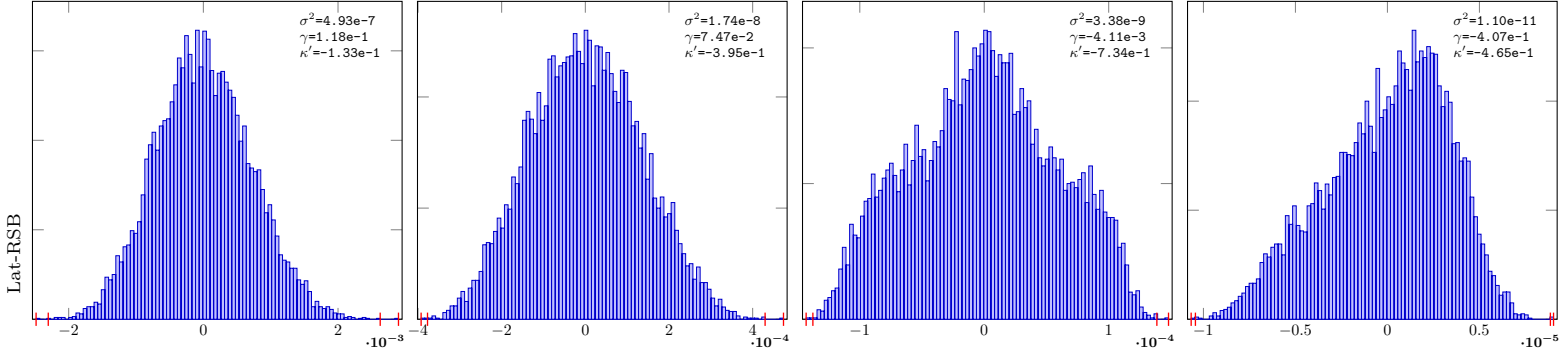


SmoothPerB4-16-Lat-RSB-10

SmoothPerB4-16-Lat-RSB-12

SmoothPerB4-16-Lat-RSB-14

SmoothPerB4-16-Lat-RSB-16

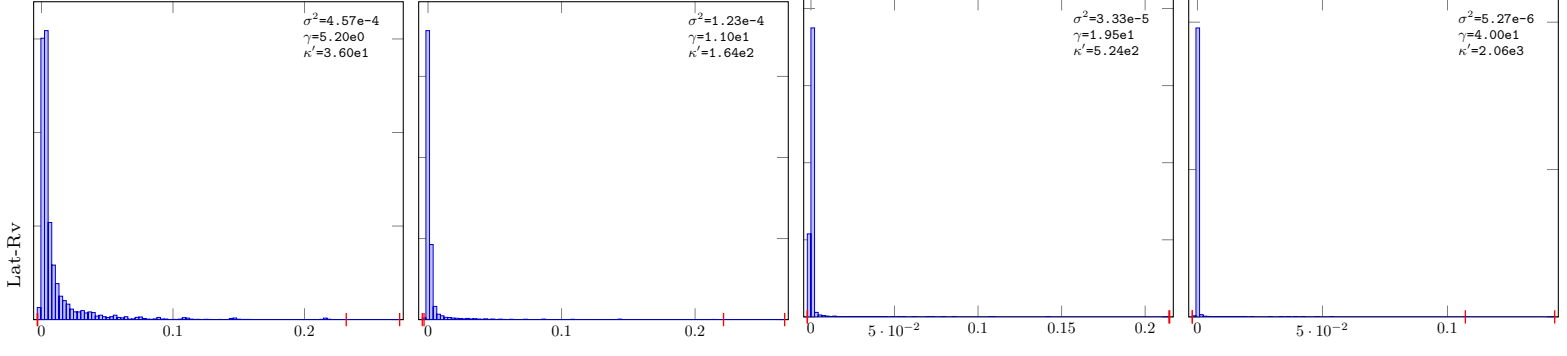


SmoothPerB4-16-Lat-Rv-10

SmoothPerB4-16-Lat-Rv-12

SmoothPerB4-16-Lat-Rv-14

SmoothPerB4-16-Lat-Rv-16

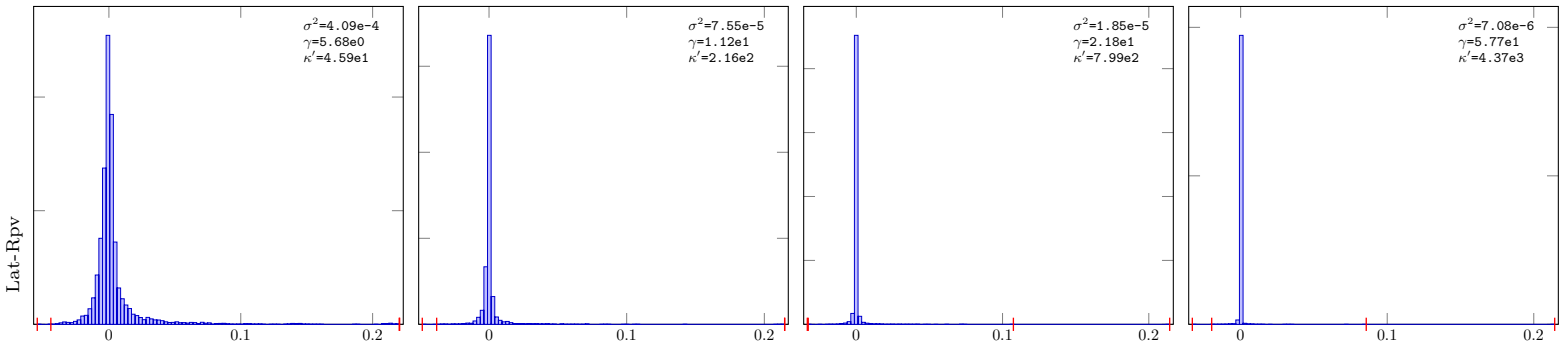


SmoothPerB4-16-Lat-Rpv-10

SmoothPerB4-16-Lat-Rpv-12

SmoothPerB4-16-Lat-Rpv-14

SmoothPerB4-16-Lat-Rpv-16



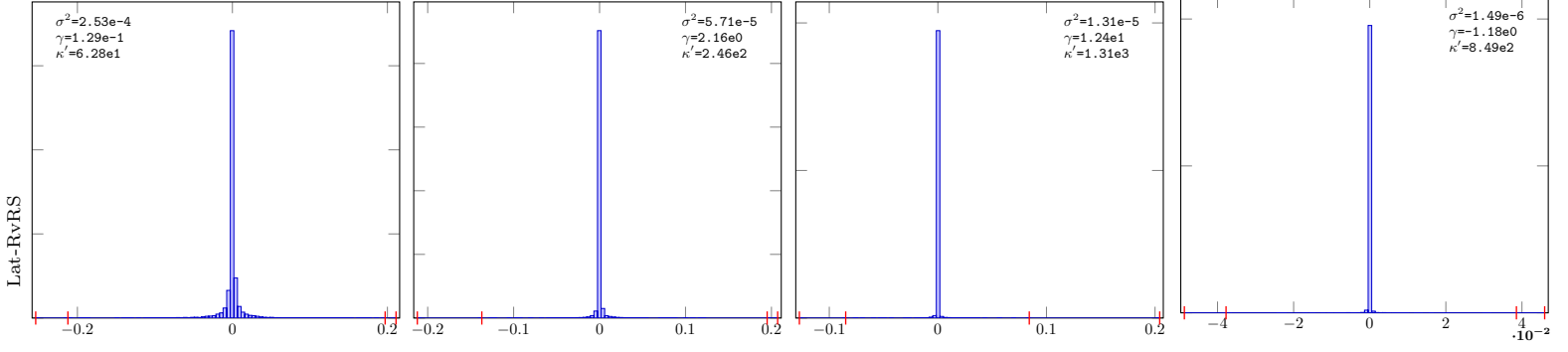
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-16-Lat-RvRS-10

SmoothPerB4-16-Lat-RvRS-12

SmoothPerB4-16-Lat-RvRS-14

SmoothPerB4-16-Lat-RvRS-16

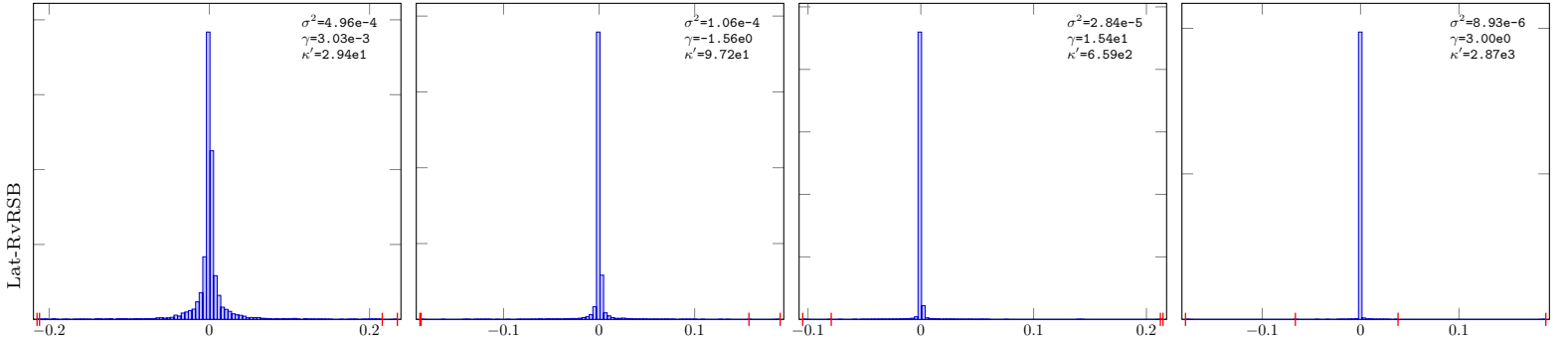


SmoothPerB4-16-Lat-RvRSB-10

SmoothPerB4-16-Lat-RvRSB-12

SmoothPerB4-16-Lat-RvRSB-14

SmoothPerB4-16-Lat-RvRSB-16

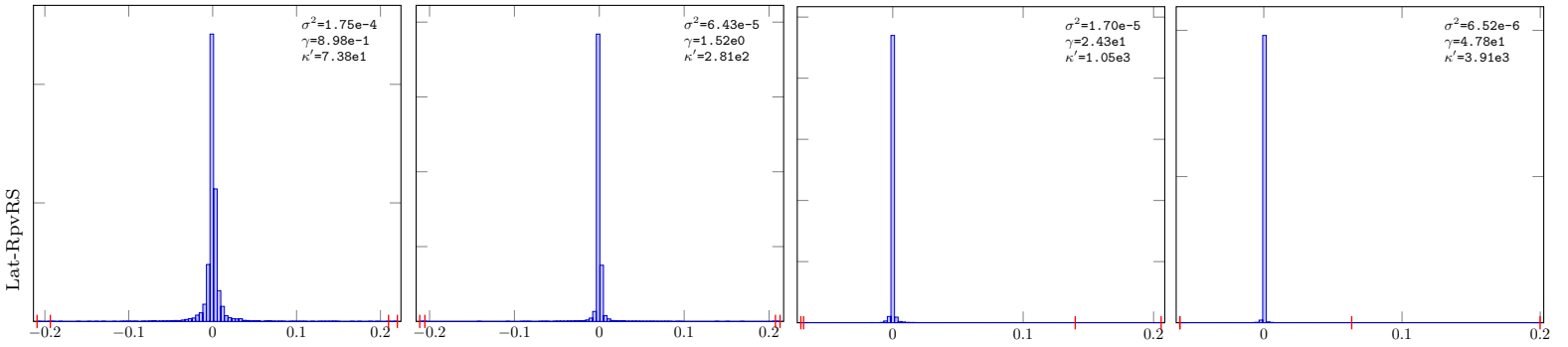


SmoothPerB4-16-Lat-RpvRS-10

SmoothPerB4-16-Lat-RpvRS-12

SmoothPerB4-16-Lat-RpvRS-14

SmoothPerB4-16-Lat-RpvRS-16

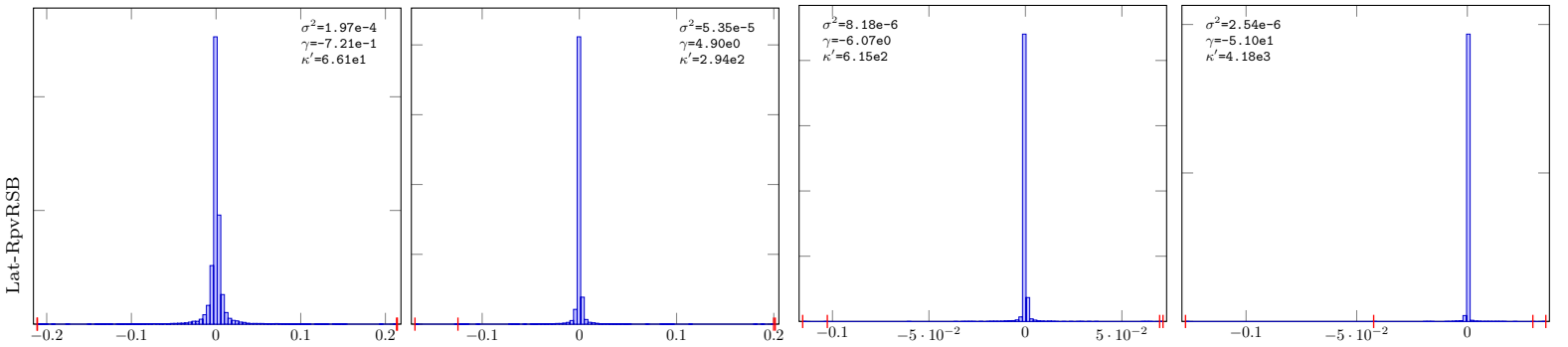


SmoothPerB4-16-Lat-RpvRSB-10

SmoothPerB4-16-Lat-RpvRSB-12

SmoothPerB4-16-Lat-RpvRSB-14

SmoothPerB4-16-Lat-RpvRSB-16



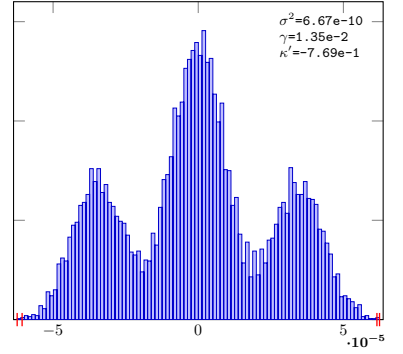
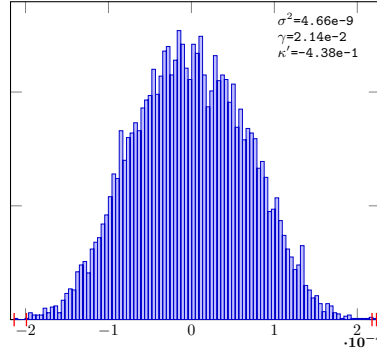
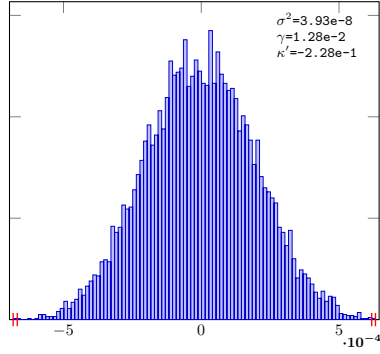
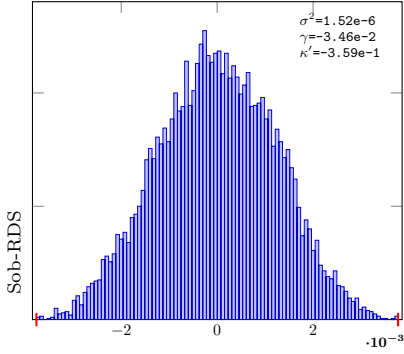
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-16-Sob-RDS-10

SmoothPerB4-16-Sob-RDS-12

SmoothPerB4-16-Sob-RDS-14

SmoothPerB4-16-Sob-RDS-16

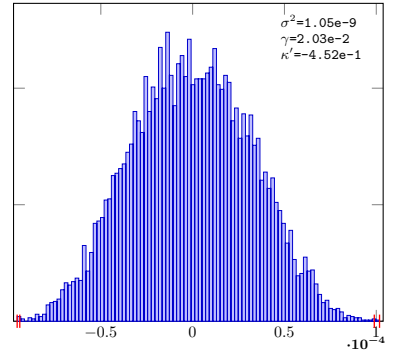
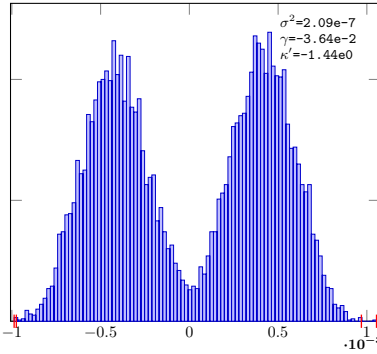
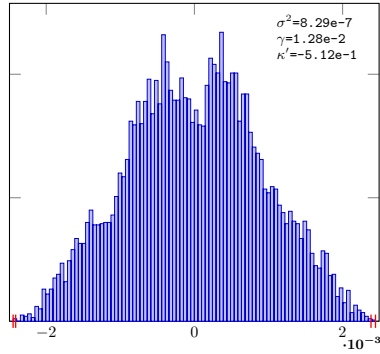
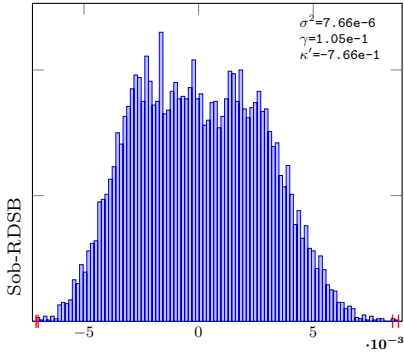


SmoothPerB4-16-Sob-RDSB-10

SmoothPerB4-16-Sob-RDSB-12

SmoothPerB4-16-Sob-RDSB-14

SmoothPerB4-16-Sob-RDSB-16

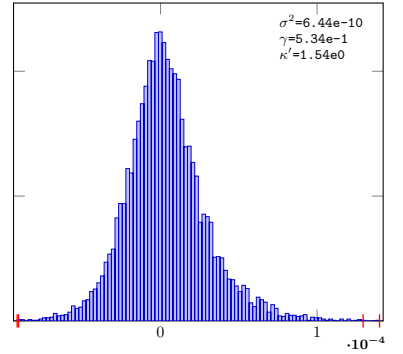
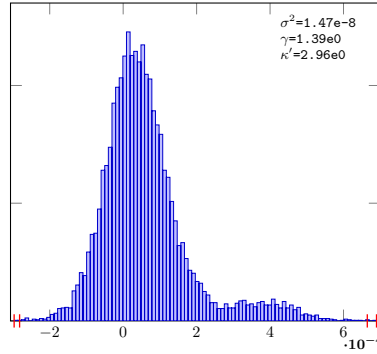
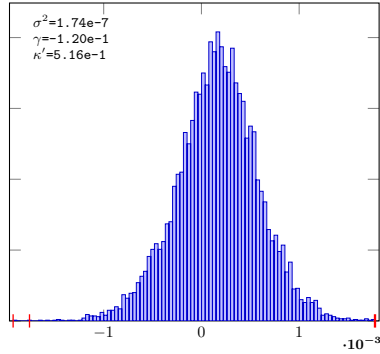
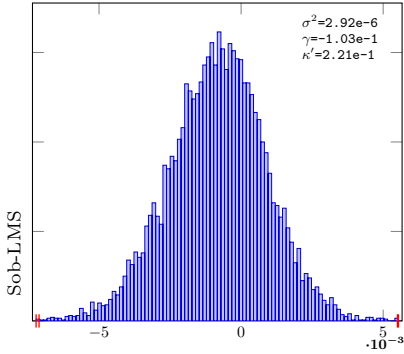


SmoothPerB4-16-Sob-LMS-10

SmoothPerB4-16-Sob-LMS-12

SmoothPerB4-16-Sob-LMS-14

SmoothPerB4-16-Sob-LMS-16

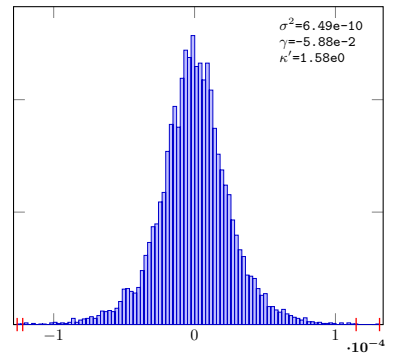
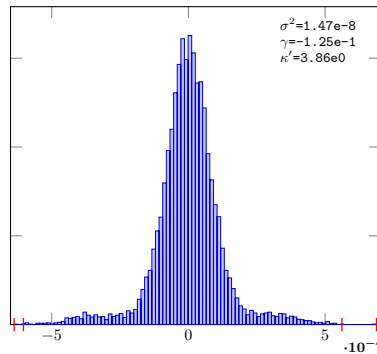
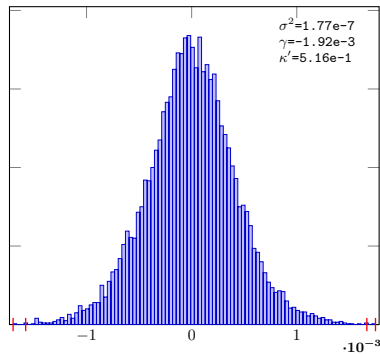
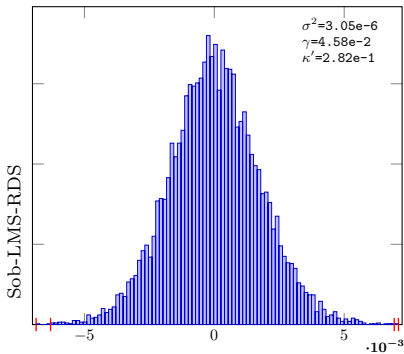


SmoothPerB4-16-Sob-LMS-RDS-10

SmoothPerB4-16-Sob-LMS-RDS-12

SmoothPerB4-16-Sob-LMS-RDS-14

SmoothPerB4-16-Sob-LMS-RDS-16



$$n = 2^{10}$$

$$n = 2^{12}$$

$$n = 2^{14}$$

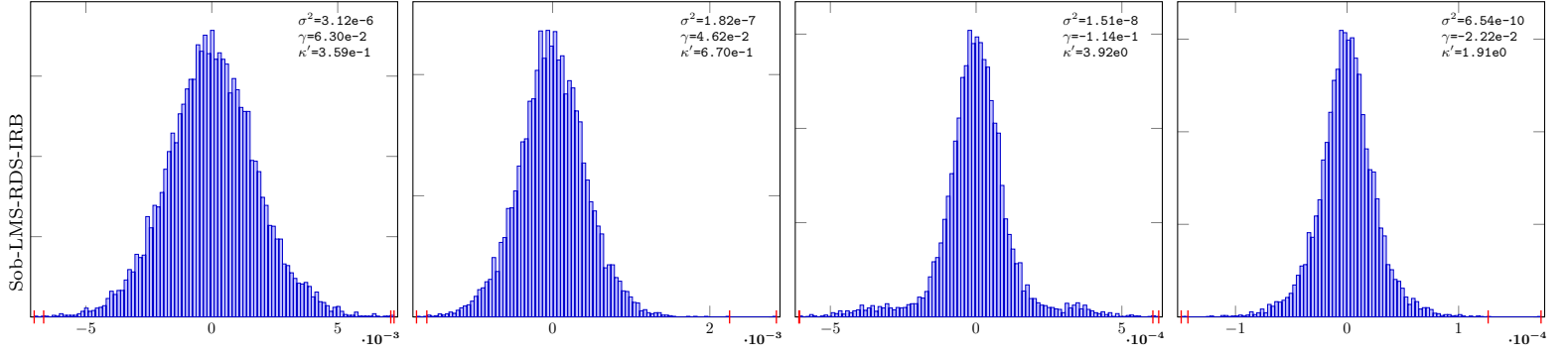
$$n = 2^{16}$$

SmoothPerB4-16-Sob-LMS-RDS-IRB-10

SmoothPerB4-16-Sob-LMS-RDS-IRB-12

SmoothPerB4-16-Sob-LMS-RDS-IRB-14

SmoothPerB4-16-Sob-LMS-RDS-IRB-16

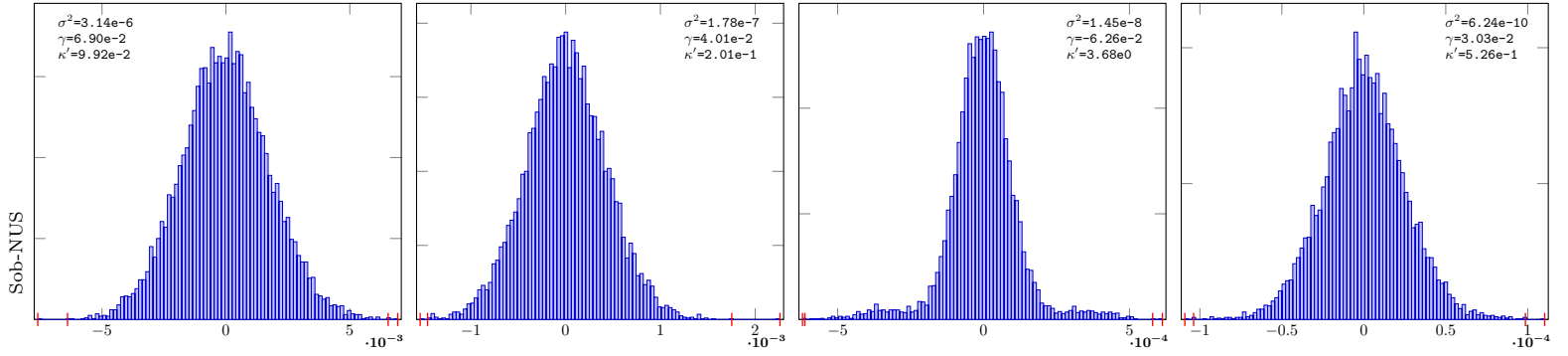


SmoothPerB4-16-Sob-NUS-10

SmoothPerB4-16-Sob-NUS-12

SmoothPerB4-16-Sob-NUS-14

SmoothPerB4-16-Sob-NUS-16



RQMC comparison: SmoothPerB4 s = 32 ( $10^4$  samples)

$n = 2^{10}$

$n = 2^{12}$

$n = 2^{14}$

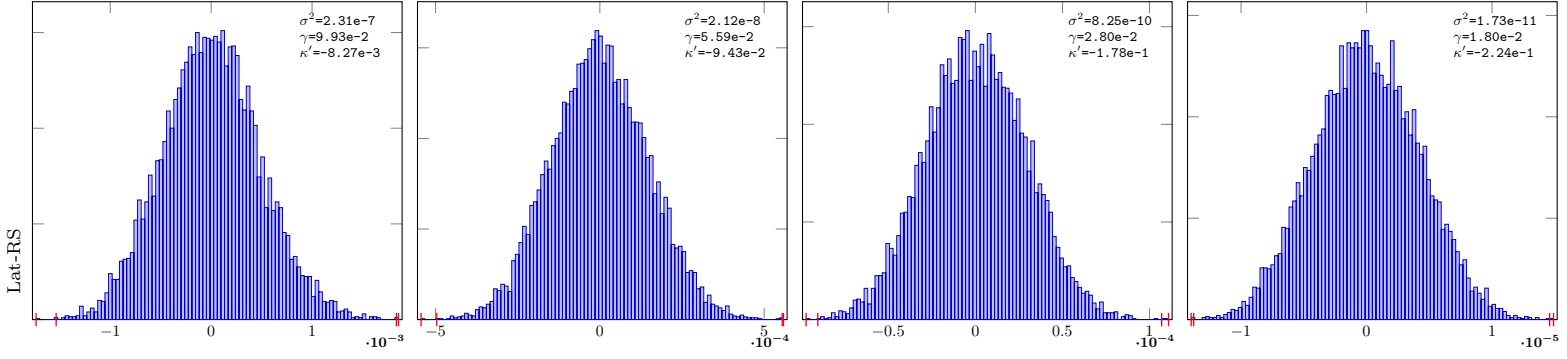
$n = 2^{16}$

SmoothPerB4-32-Lat-RS-10

SmoothPerB4-32-Lat-RS-12

SmoothPerB4-32-Lat-RS-14

SmoothPerB4-32-Lat-RS-16

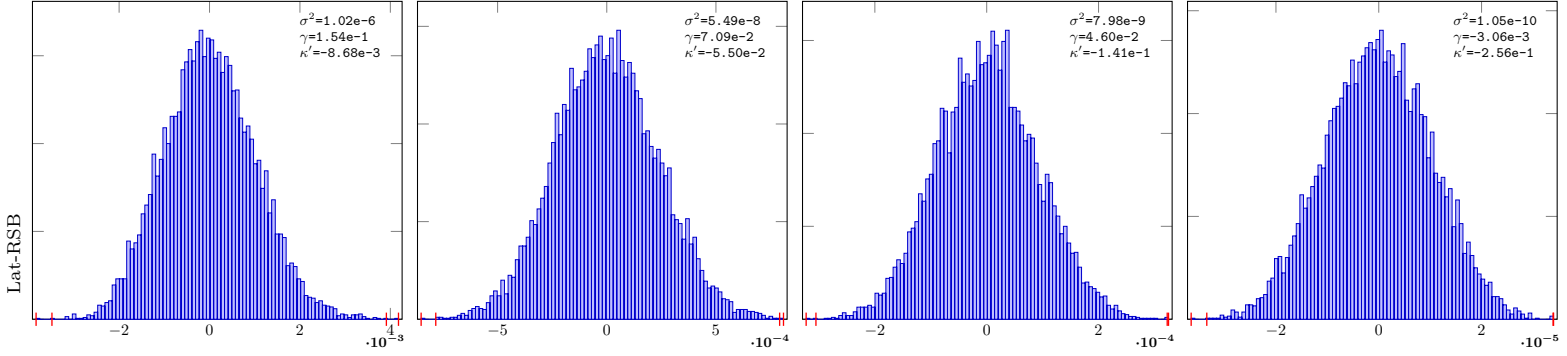


SmoothPerB4-32-Lat-RSB-10

SmoothPerB4-32-Lat-RSB-12

SmoothPerB4-32-Lat-RSB-14

SmoothPerB4-32-Lat-RSB-16

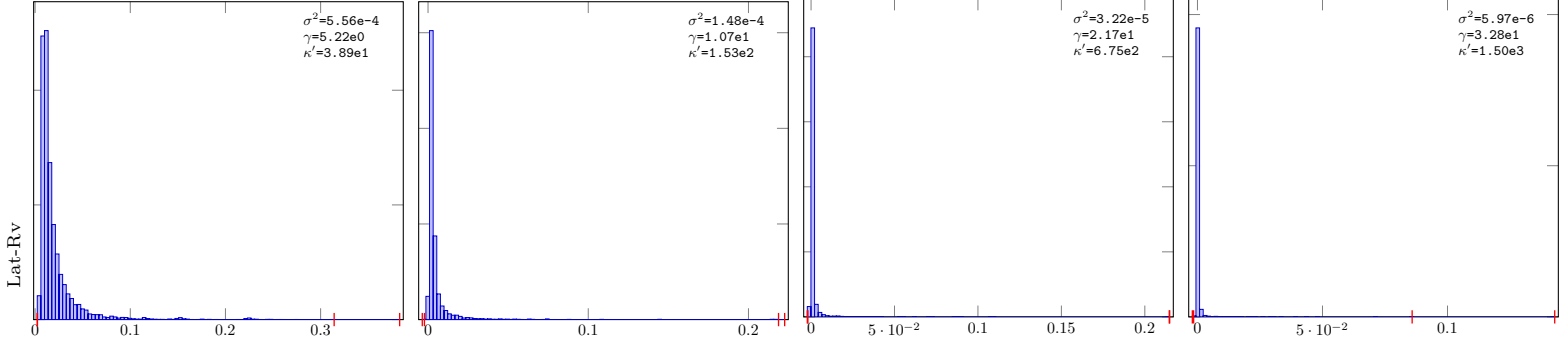


SmoothPerB4-32-Lat-Rv-10

SmoothPerB4-32-Lat-Rv-12

SmoothPerB4-32-Lat-Rv-14

SmoothPerB4-32-Lat-Rv-16

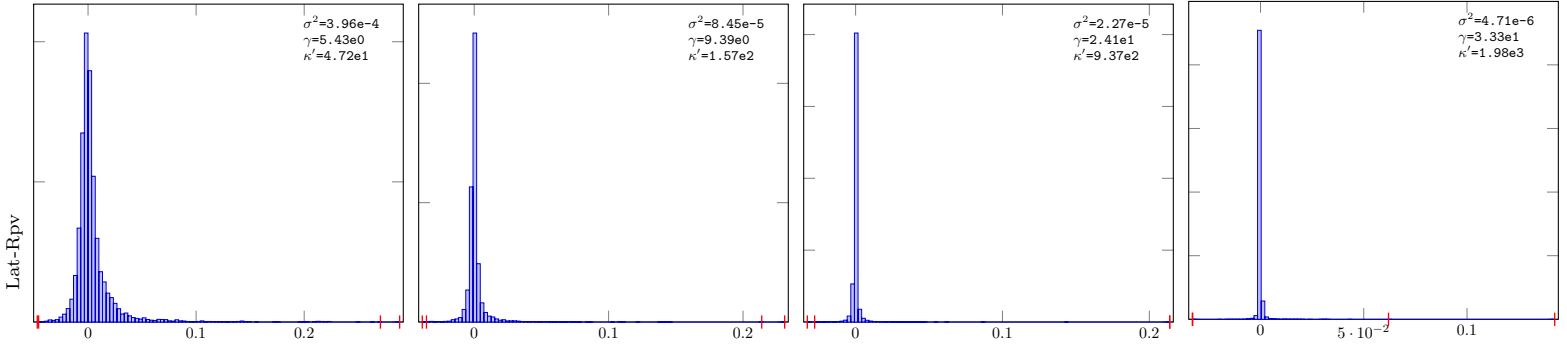


SmoothPerB4-32-Lat-Rpv-10

SmoothPerB4-32-Lat-Rpv-12

SmoothPerB4-32-Lat-Rpv-14

SmoothPerB4-32-Lat-Rpv-16



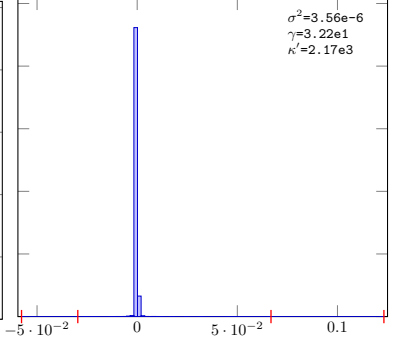
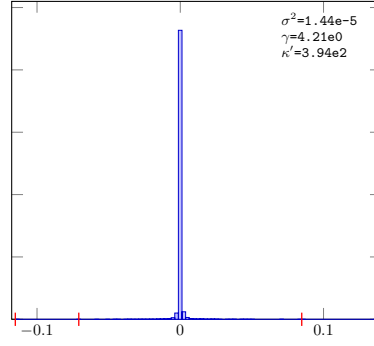
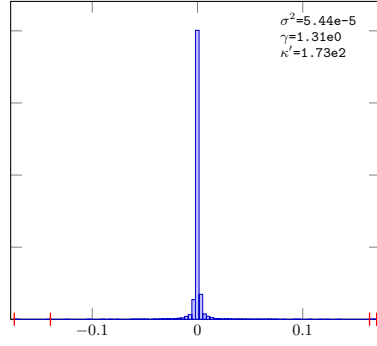
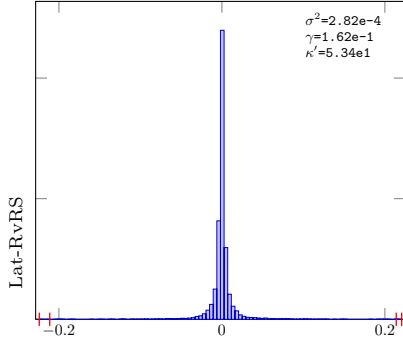
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-32-Lat-RvRS-10

SmoothPerB4-32-Lat-RvRS-12

SmoothPerB4-32-Lat-RvRS-14

SmoothPerB4-32-Lat-RvRS-16

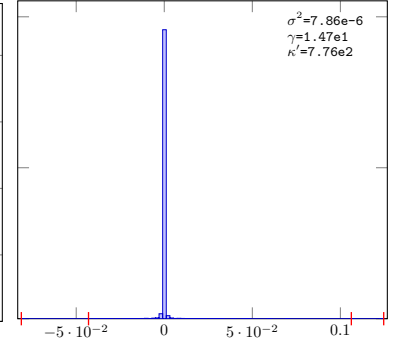
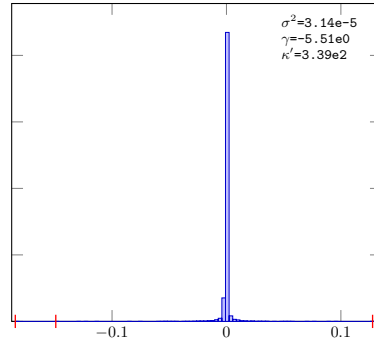
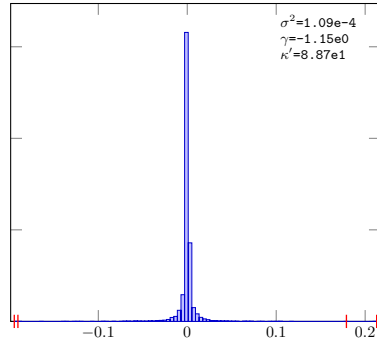
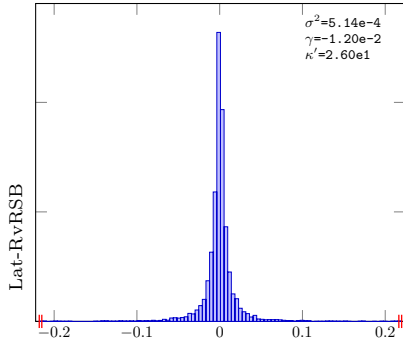


SmoothPerB4-32-Lat-RvRSB-10

SmoothPerB4-32-Lat-RvRSB-12

SmoothPerB4-32-Lat-RvRSB-14

SmoothPerB4-32-Lat-RvRSB-16

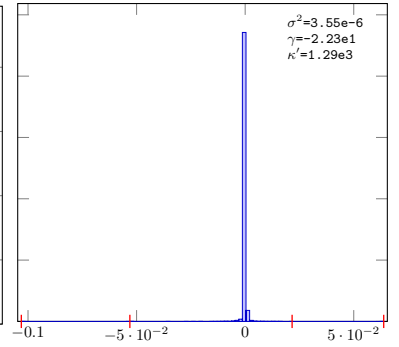
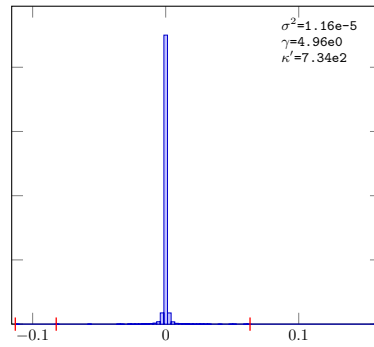
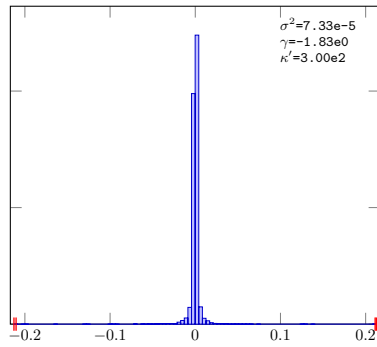
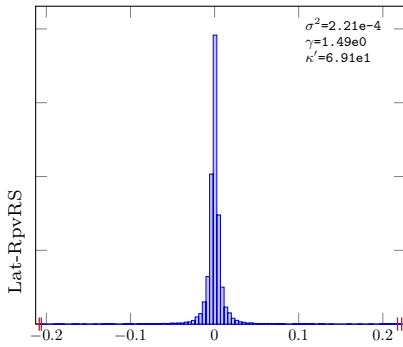


SmoothPerB4-32-Lat-RpvRS-10

SmoothPerB4-32-Lat-RpvRS-12

SmoothPerB4-32-Lat-RpvRS-14

SmoothPerB4-32-Lat-RpvRS-16

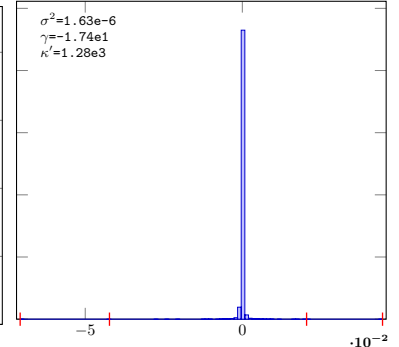
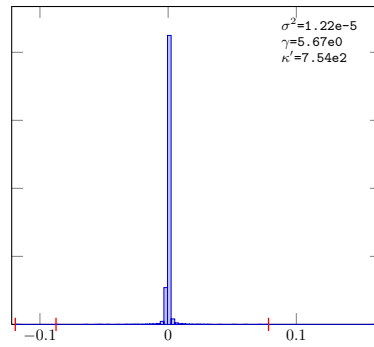
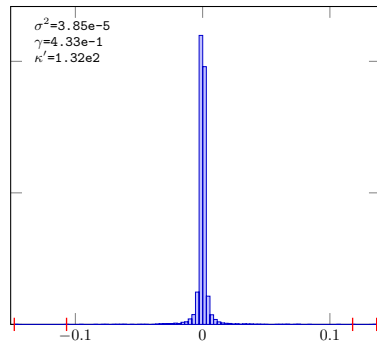
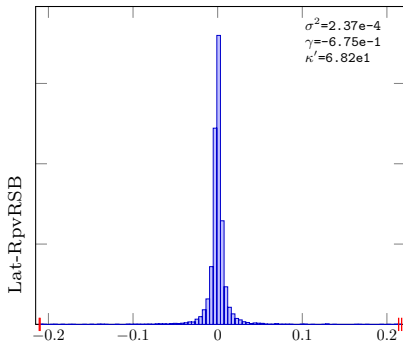


SmoothPerB4-32-Lat-RpvRSB-10

SmoothPerB4-32-Lat-RpvRSB-12

SmoothPerB4-32-Lat-RpvRSB-14

SmoothPerB4-32-Lat-RpvRSB-16



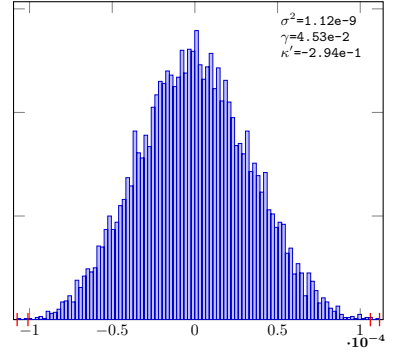
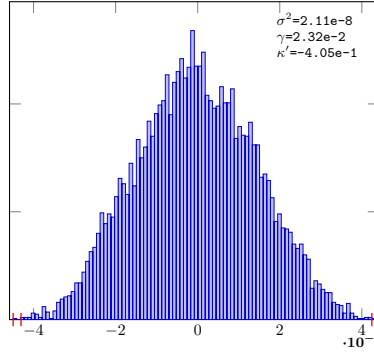
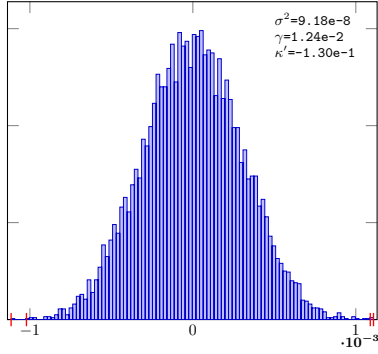
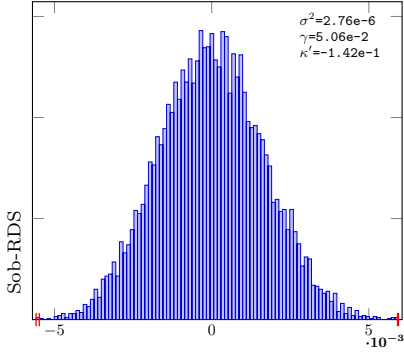
$n = 2^{10}$  $n = 2^{12}$  $n = 2^{14}$  $n = 2^{16}$ 

SmoothPerB4-32-Sob-RDS-10

SmoothPerB4-32-Sob-RDS-12

SmoothPerB4-32-Sob-RDS-14

SmoothPerB4-32-Sob-RDS-16

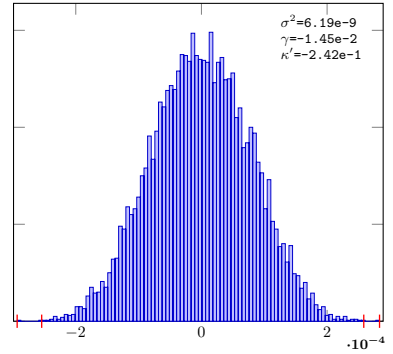
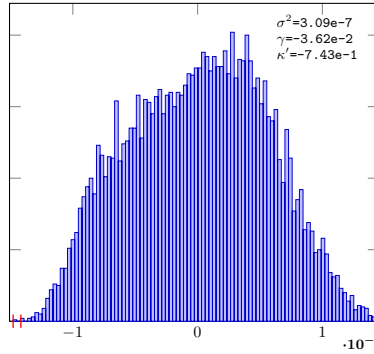
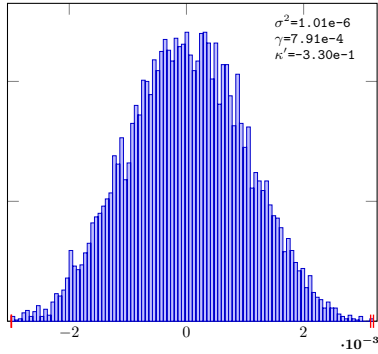
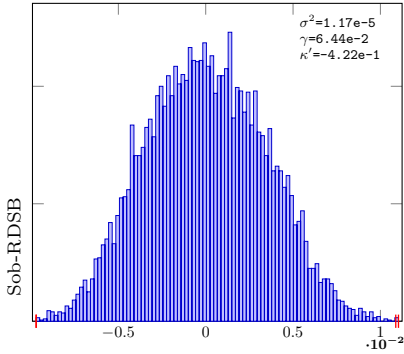


SmoothPerB4-32-Sob-RDSB-10

SmoothPerB4-32-Sob-RDSB-12

SmoothPerB4-32-Sob-RDSB-14

SmoothPerB4-32-Sob-RDSB-16

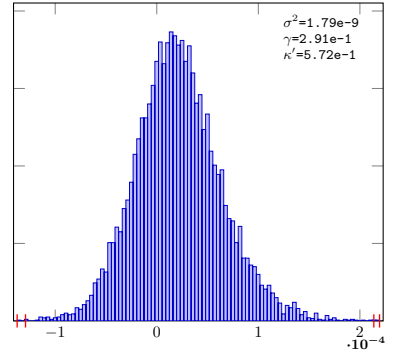
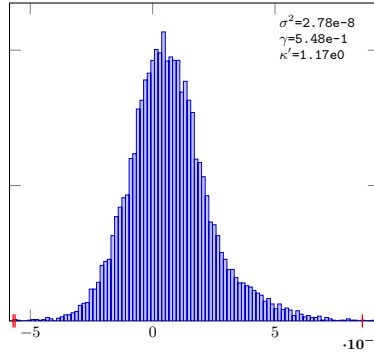
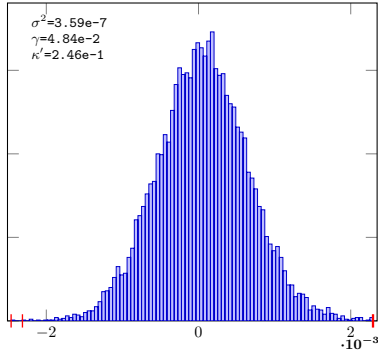
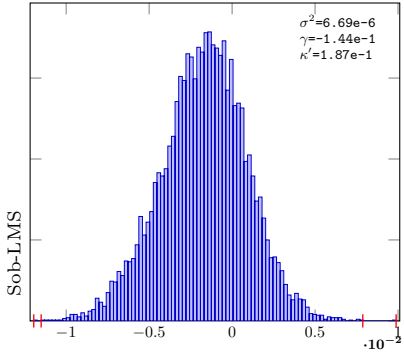


SmoothPerB4-32-Sob-LMS-10

SmoothPerB4-32-Sob-LMS-12

SmoothPerB4-32-Sob-LMS-14

SmoothPerB4-32-Sob-LMS-16

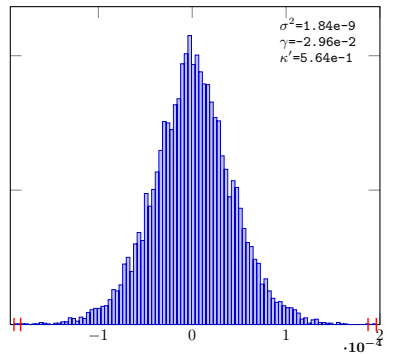
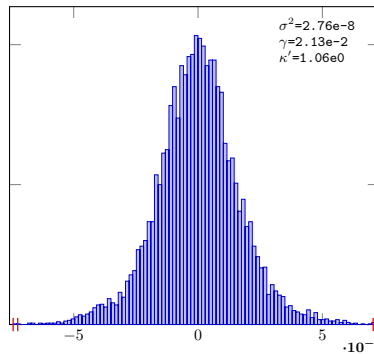
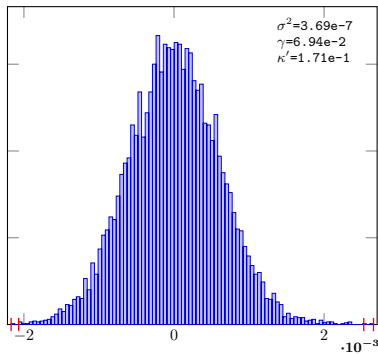
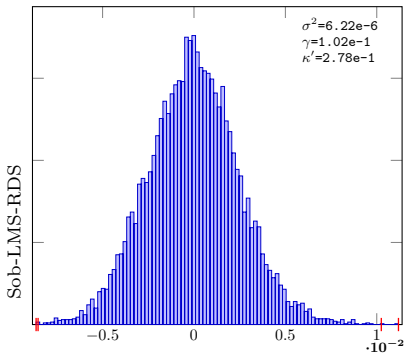


SmoothPerB4-32-Sob-LMS-RDS-10

SmoothPerB4-32-Sob-LMS-RDS-12

SmoothPerB4-32-Sob-LMS-RDS-14

SmoothPerB4-32-Sob-LMS-RDS-16



$$n = 2^{10}$$

$$n = 2^{12}$$

$$n = 2^{14}$$

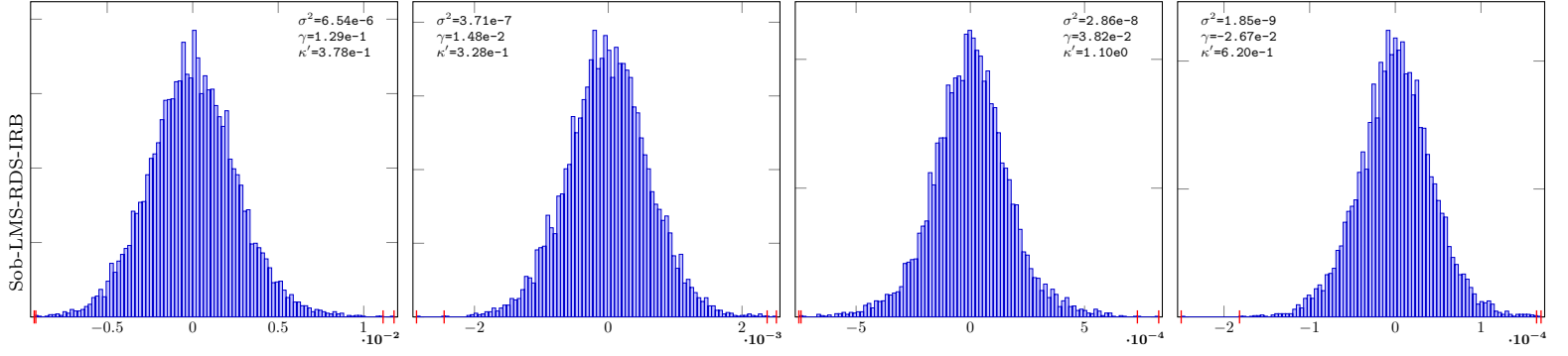
$$n = 2^{16}$$

SmoothPerB4-32-Sob-LMS-RDS-IRB-10

SmoothPerB4-32-Sob-LMS-RDS-IRB-12

SmoothPerB4-32-Sob-LMS-RDS-IRB-14

SmoothPerB4-32-Sob-LMS-RDS-IRB-16



SmoothPerB4-32-Sob-NUS-10

SmoothPerB4-32-Sob-NUS-12

SmoothPerB4-32-Sob-NUS-14

SmoothPerB4-32-Sob-NUS-16

